

```

-> 63 targets ..... [14.9s, 2+, 15-]

.. Derive: OsteoLaus x 20 imputed datasets .....
-> 63 targets ..... [14.9s, 2+, 15-]
[observed/20] deriving ...

-> 63 targets ..... [14.9s, 2+, 15-]

.. Derive BMI ..
-> 63 targets ..... [14.9s, 2+, 15-]

-> 63 targets ..... [14.9s, 2+, 15-]
. BMI derived from Weight and Height.
. Total rows: 6114 | derived: 6113 | missing: 1
. BMI range (non-missing): 15.1 - 45.5 kg/m2
. Mean ± SD: 25.9 ± 4.7 kg/m2
. Agreement with BMI_source (rows with both non-missing, n = 6112):
. Difference < 0.01 (effectively identical): 6099 (99.8%)
. Difference 0.01-0.5 (minor rounding): 13 (0.2%)
. Difference >= 0.5 (meaningful mismatch): 0 (0%)
. Median absolute difference: 0.002 kg/m2
. Max absolute difference : 0.17 kg/m2

-> 63 targets ..... [14.9s, 2+, 15-]

.. Deriving BMI category ..
-> 63 targets ..... [14.9s, 2+, 15-]

-> 63 targets ..... [14.9s, 2+, 15-]
. derive_bmi_category: BMI categories derived.
. Processed 6114 rows (1 missing BMI).
  Prevalence of Obesity: 18.5%
  Prevalence of Overweight + Obesity: 53.4%

-> 63 targets ..... [14.9s, 2+, 15-]

.. Derive ALM Indices ..
-> 63 targets ..... [14.9s, 2+, 15-]

-> 63 targets ..... [14.9s, 2+, 15-]

.. Raw indices (native DXA platform)
-> 63 targets ..... [14.9s, 2+, 15-]
. 'ALM': 5635 derived, 479 missing
. Range 9061.706 - 27682.402 |
  Mean ± SD: 17024.849 ± 2595.696
. 'ALM_HT2': 5635 derived, 479 missing
. Range 4.1 - 10.943 |
  Mean ± SD: 6.581 ± 0.86
. 'ALM_BMI': 5635 derived, 479 missing

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. Range 0.379 - 1.054 |
  Mean ± SD: 0.668 ± 0.101
. 'ALM_WT': 5635 derived, 479 missing
. Range 0.178 - 0.381 |
  Mean ± SD: 0.258 ± 0.028
```

```
-> 63 targets ..... [14.9s, 2+, 15-]
```

```
.. Harmonised indices (Lunar converted to Hologic scale)
-> 63 targets ..... [14.9s, 2+, 15-]
. 'ALM_harmonised': 5635 derived, 479 missing
. Range 9061.706 - 27682.402 |
  Mean ± SD: 16487.142 ± 2625.595
. 'ALM_HT2_harmonised': 5635 derived, 479 missing
. Range 3.811 - 10.943 |
  Mean ± SD: 6.371 ± 0.874
. 'ALM_BMI_harmonised': 5635 derived, 479 missing
. Range 0.354 - 1.054 |
  Mean ± SD: 0.647 ± 0.102
. 'ALM_WT_harmonised': 5635 derived, 479 missing
. Range 0.168 - 0.381 |
  Mean ± SD: 0.249 ± 0.029
. Sanity check passed: ALM_harmonised == ALM for all 2360 Hologic rows.
. Agreement 'ALM' vs 'ALM_source' (n = 5635):
. < 0.001: 5629 (99.9%)
. 0.001 to 0.1: 0 (0%)
. >= 0.1: 6 (0.1%)
. Median |diff|: 0 | Max: 9.0683
6 row(s) differ by >= 0.1 between derived and source.
. Agreement 'ALM_HT2' vs 'ALM_HT2_source' (n = 5635):
. < 0.001: 5629 (99.9%)
. 0.001 to 0.1: 6 (0.1%)
. >= 0.1: 0 (0%)
. Median |diff|: 0 | Max: 0.0035
. Agreement 'ALM_BMI' vs 'ALM_BMI_source' (n = 5634):
. < 0.001: 5629 (99.9%)
. 0.001 to 0.1: 5 (0.1%)
. >= 0.1: 0 (0%)
. Median |diff|: 1e-04 | Max: 0.0052
. Agreement 'ALM_WT' vs 'ALM_WT_source' (n = 5635):
. < 0.001: 5635 (100%)
. 0.001 to 0.1: 0 (0%)
. >= 0.1: 0 (0%)
. Median |diff|: 0 | Max: 2e-04
```

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-> 63 targets ..... [14.9s, 2+, 15-]
```

```
.. Split ALM by DXA method ..
```

```
-> 63 targets ..... [14.9s, 2+, 15-]
```

```
-> 63 targets ..... [14.9s, 2+, 15-]
. Method-specific ALM columns created:
. Hologic and Lunar splits for ALM, ALM_HT2, ALM_BMI, ALM_WT
```

```

-> 63 targets ..... [14.9s, 2+, 15-]

.. Derive Age ..
-> 63 targets ..... [14.9s, 2+, 15-]

-> 63 targets ..... [14.9s, 2+, 15-]
. Age derived from datbirth (Baseline, filled forward/back) and exam_date_iso.
. Total rows: 6114 | derived: 6114 | missing: 0
. Age range (non-missing): 50.2 - 91.5 years
. Mean ± SD: 68.6 ± 7.9 years
. Agreement with Age_source (rows with both non-missing, n = 6114):
. Difference < 0.1 yr (effectively identical): 6066 (99.2%)
. Difference 0.1-1 yr (minor rounding/timing): 47 (0.8%)
. Difference >= 1 yr (meaningful mismatch) : 1 (0%)
. Median absolute difference: 0.028 years
. Max absolute difference : 4.328 years
[1/20] deriving ...

-> 63 targets ..... [14.9s, 2+, 15-]

.. Derive BMI ..
-> 63 targets ..... [14.9s, 2+, 15-]

-> 63 targets ..... [14.9s, 2+, 15-]
. BMI derived from Weight and Height.
. Total rows: 6114 | derived: 6114 | missing: 0
. BMI range (non-missing): 15.1 - 45.5 kg/m²
. Mean ± SD: 25.9 ± 4.7 kg/m²
. Agreement with BMI_source (rows with both non-missing, n = 6112):
. Difference < 0.01 (effectively identical): 6099 (99.8%)
. Difference 0.01-0.5 (minor rounding): 13 (0.2%)
. Difference >= 0.5 (meaningful mismatch): 0 (0%)
. Median absolute difference: 0.002 kg/m²
. Max absolute difference : 0.17 kg/m²

-> 63 targets ..... [14.9s, 2+, 15-]

.. Deriving BMI category ..
-> 63 targets ..... [14.9s, 2+, 15-]

-> 63 targets ..... [14.9s, 2+, 15-]
. derive_bmi_category: BMI categories derived.
. Processed 6114 rows (0 missing BMI).
  Prevalence of Obesity: 18.5%
  Prevalence of Overweight + Obesity: 53.4%

-> 63 targets ..... [14.9s, 2+, 15-]

.. Derive ALM Indices ..
-> 63 targets ..... [14.9s, 2+, 15-]

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-> 63 targets ..... [14.9s, 2+, 15-]

.. Raw indices (native DXA platform)
-> 63 targets ..... [14.9s, 2+, 15-]
. 'ALM': 6114 derived, 0 missing
. Range 9061.706 - 27682.402 |
  Mean ± SD: 17007.617 ± 2588.005
. 'ALM_HT2': 6114 derived, 0 missing
. Range 4.1 - 10.943 |
  Mean ± SD: 6.574 ± 0.859
. 'ALM_BMI': 6114 derived, 0 missing
. Range 0.379 - 1.054 |
  Mean ± SD: 0.666 ± 0.101
. 'ALM_WT': 6114 derived, 0 missing
. Range 0.178 - 0.381 |
  Mean ± SD: 0.257 ± 0.028

-> 63 targets ..... [14.9s, 2+, 15-]

.. Harmonised indices (Lunar converted to Hologic scale)
-> 63 targets ..... [14.9s, 2+, 15-]
. 'ALM_harmonised': 6114 derived, 0 missing
. Range 9061.706 - 27682.402 |
  Mean ± SD: 16509.804 ± 2616.802
. 'ALM_HT2_harmonised': 6114 derived, 0 missing
. Range 3.811 - 10.943 |
  Mean ± SD: 6.38 ± 0.872
. 'ALM_BMI_harmonised': 6114 derived, 0 missing
. Range 0.354 - 1.054 |
  Mean ± SD: 0.647 ± 0.101
. 'ALM_WT_harmonised': 6114 derived, 0 missing
. Range 0.168 - 0.381 |
  Mean ± SD: 0.249 ± 0.028
. Sanity check passed: ALM_harmonised == ALM for all 2824 Hologic rows.
. Agreement 'ALM' vs 'ALM_source' (n = 5636):
. < 0.001: 5629 (99.9%)
. 0.001 to 0.1: 0 (0%)
. >= 0.1: 7 (0.1%)
. Median |diff|: 0 | Max: 417.0114
7 row(s) differ by >= 0.1 between derived and source.
. Agreement 'ALM_HT2' vs 'ALM_HT2_source' (n = 5636):
. < 0.001: 5629 (99.9%)
. 0.001 to 0.1: 6 (0.1%)
. >= 0.1: 1 (0%)
. Median |diff|: 0 | Max: 0.1681
1 row(s) differ by >= 0.1 between derived and source.
. Agreement 'ALM_BMI' vs 'ALM_BMI_source' (n = 5635):
. < 0.001: 5629 (99.9%)
. 0.001 to 0.1: 6 (0.1%)
. >= 0.1: 0 (0%)
. Median |diff|: 1e-04 | Max: 0.0182
. Agreement 'ALM_WT' vs 'ALM_WT_source' (n = 5636):
. < 0.001: 5635 (100%)
. 0.001 to 0.1: 1 (0%)
. >= 0.1: 0 (0%)
. Median |diff|: 0 | Max: 0.0074

```

```

-> 63 targets ..... [14.9s, 2+, 15-]

.. Split ALM by DXA method ..
-> 63 targets ..... [14.9s, 2+, 15-]

-> 63 targets ..... [14.9s, 2+, 15-]
. Method-specific ALM columns created:
. Hologic and Lunar splits for ALM, ALM_HT2, ALM_BMI, ALM_WT

-> 63 targets ..... [14.9s, 2+, 15-]

.. Derive Age ..
-> 63 targets ..... [14.9s, 2+, 15-]

-> 63 targets ..... [14.9s, 2+, 15-]
. Age derived from datbirth (Baseline, filled forward/back) and exam_date_iso.
. Total rows: 6114 | derived: 6114 | missing: 0
. Age range (non-missing): 50.2 - 91.5 years
. Mean ± SD: 68.6 ± 7.9 years
. Agreement with Age_source (rows with both non-missing, n = 6114):
. Difference < 0.1 yr (effectively identical): 6066 (99.2%)
. Difference 0.1-1 yr (minor rounding/timing): 47 (0.8%)
. Difference >= 1 yr (meaningful mismatch) : 1 (0%)
. Median absolute difference: 0.028 years
. Max absolute difference : 4.328 years
[2/20] deriving ...

-> 63 targets ..... [14.9s, 2+, 15-]

.. Derive BMI ..
-> 63 targets ..... [14.9s, 2+, 15-]

-> 63 targets ..... [14.9s, 2+, 15-]
. BMI derived from Weight and Height.
. Total rows: 6114 | derived: 6114 | missing: 0
. BMI range (non-missing): 15.1 - 45.5 kg/m²
. Mean ± SD: 25.9 ± 4.7 kg/m²
. Agreement with BMI_source (rows with both non-missing, n = 6112):
. Difference < 0.01 (effectively identical): 6099 (99.8%)
. Difference 0.01-0.5 (minor rounding): 13 (0.2%)
. Difference >= 0.5 (meaningful mismatch): 0 (0%)
. Median absolute difference: 0.002 kg/m²
. Max absolute difference : 0.17 kg/m²

-> 63 targets ..... [14.9s, 2+, 15-]

.. Deriving BMI category ..
-> 63 targets ..... [14.9s, 2+, 15-]

-> 63 targets ..... [14.9s, 2+, 15-]

```

```

. derive_bmi_category: BMI categories derived.
. Processed 6114 rows (0 missing BMI).
  Prevalence of Obesity: 18.5%
  Prevalence of Overweight + Obesity: 53.4%

-> 63 targets ..... [14.9s, 2+, 15-]

.. Derive ALM Indices ..
-> 63 targets ..... [14.9s, 2+, 15-]

-> 63 targets ..... [14.9s, 2+, 15-]

.. Raw indices (native DXA platform)
-> 63 targets ..... [14.9s, 2+, 15-]
. 'ALM': 6114 derived, 0 missing
. Range 9061.706 - 27682.402 |
  Mean ± SD: 17006.243 ± 2587.159
. 'ALM_HT2': 6114 derived, 0 missing
. Range 4.1 - 10.943 |
  Mean ± SD: 6.573 ± 0.857
. 'ALM_BMI': 6114 derived, 0 missing
. Range 0.379 - 1.054 |
  Mean ± SD: 0.666 ± 0.101
. 'ALM_WT': 6114 derived, 0 missing
. Range 0.178 - 0.381 |
  Mean ± SD: 0.257 ± 0.028

-> 63 targets ..... [14.9s, 2+, 15-]

.. Harmonised indices (Lunar converted to Hologic scale)
-> 63 targets ..... [14.9s, 2+, 15-]
. 'ALM_harmonised': 6114 derived, 0 missing
. Range 9061.706 - 27682.402 |
  Mean ± SD: 16508.437 ± 2615.754
. 'ALM_HT2_harmonised': 6114 derived, 0 missing
. Range 3.811 - 10.943 |
  Mean ± SD: 6.38 ± 0.87
. 'ALM_BMI_harmonised': 6114 derived, 0 missing
. Range 0.354 - 1.054 |
  Mean ± SD: 0.647 ± 0.102
. 'ALM_WT_harmonised': 6114 derived, 0 missing
. Range 0.168 - 0.381 |
  Mean ± SD: 0.249 ± 0.028
. Sanity check passed: ALM_harmonised == ALM for all 2824 Hologic rows.
. Agreement 'ALM' vs 'ALM_source' (n = 5636):
. < 0.001: 5629 (99.9%)
. 0.001 to 0.1: 0 (0%)
. >= 0.1: 7 (0.1%)
. Median |diff|: 0 | Max: 181.7815
7 row(s) differ by >= 0.1 between derived and source.
. Agreement 'ALM_HT2' vs 'ALM_HT2_source' (n = 5636):
. < 0.001: 5629 (99.9%)
. 0.001 to 0.1: 7 (0.1%)
. >= 0.1: 0 (0%)
. Median |diff|: 0 | Max: 0.0733

```

```

. Agreement 'ALM_BMI' vs 'ALM_BMI_source' (n = 5635):
. < 0.001: 5629 (99.9%)
. 0.001 to 0.1: 6 (0.1%)
. >= 0.1: 0 (0%)
. Median |diff|: 1e-04 | Max: 0.0079
. Agreement 'ALM_WT' vs 'ALM_WT_source' (n = 5636):
. < 0.001: 5635 (100%)
. 0.001 to 0.1: 1 (0%)
. >= 0.1: 0 (0%)
. Median |diff|: 0 | Max: 0.0032

-> 63 targets ..... [14.9s, 2+, 15-]

.. Split ALM by DXA method ..
-> 63 targets ..... [14.9s, 2+, 15-]

-> 63 targets ..... [14.9s, 2+, 15-]
. Method-specific ALM columns created:
. Hologic and Lunar splits for ALM, ALM_HT2, ALM_BMI, ALM_WT

-> 63 targets ..... [14.9s, 2+, 15-]

.. Derive Age ..
-> 63 targets ..... [14.9s, 2+, 15-]

-> 63 targets ..... [14.9s, 2+, 15-]
. Age derived from datbirth (Baseline, filled forward/back) and exam_date_iso.
. Total rows: 6114 | derived: 6114 | missing: 0
. Age range (non-missing): 50.2 - 91.5 years
. Mean ± SD: 68.6 ± 7.9 years
. Agreement with Age_source (rows with both non-missing, n = 6114):
. Difference < 0.1 yr (effectively identical): 6066 (99.2%)
. Difference 0.1-1 yr (minor rounding/timing): 47 (0.8%)
. Difference >= 1 yr (meaningful mismatch) : 1 (0%)
. Median absolute difference: 0.028 years
. Max absolute difference : 4.328 years
[3/20] deriving ...

-> 63 targets ..... [14.9s, 2+, 15-]

.. Derive BMI ..
-> 63 targets ..... [14.9s, 2+, 15-]

-> 63 targets ..... [14.9s, 2+, 15-]
. BMI derived from Weight and Height.
. Total rows: 6114 | derived: 6114 | missing: 0
. BMI range (non-missing): 15.1 - 45.5 kg/m²
. Mean ± SD: 25.9 ± 4.7 kg/m²
. Agreement with BMI_source (rows with both non-missing, n = 6112):
. Difference < 0.01 (effectively identical): 6099 (99.8%)
. Difference 0.01-0.5 (minor rounding): 13 (0.2%)
. Difference >= 0.5 (meaningful mismatch): 0 (0%)

```

```

. Median absolute difference: 0.002 kg/m²
. Max absolute difference : 0.17 kg/m²

-> 63 targets ..... [14.9s, 2+, 15-]

.. Deriving BMI category ..
-> 63 targets ..... [14.9s, 2+, 15-]

-> 63 targets ..... [14.9s, 2+, 15-]
. derive_bmi_category: BMI categories derived.
. Processed 6114 rows (0 missing BMI).
  Prevalence of Obesity: 18.5%
  Prevalence of Overweight + Obesity: 53.4%

-> 63 targets ..... [14.9s, 2+, 15-]

.. Derive ALM Indices ..
-> 63 targets ..... [14.9s, 2+, 15-]

-> 63 targets ..... [14.9s, 2+, 15-]

.. Raw indices (native DXA platform)
-> 63 targets ..... [14.9s, 2+, 15-]
. 'ALM': 6114 derived, 0 missing
. Range 9061.706 - 27682.402 |
  Mean ± SD: 16999.435 ± 2587.343
. 'ALM_HT2': 6114 derived, 0 missing
. Range 4.1 - 10.943 |
  Mean ± SD: 6.571 ± 0.858
. 'ALM_BMI': 6114 derived, 0 missing
. Range 0.357 - 1.054 |
  Mean ± SD: 0.666 ± 0.101
. 'ALM_WT': 6114 derived, 0 missing
. Range 0.165 - 0.381 |
  Mean ± SD: 0.257 ± 0.028

-> 63 targets ..... [14.9s, 2+, 15-]

.. Harmonised indices (Lunar converted to Hologic scale)
-> 63 targets ..... [14.9s, 2+, 15-]
. 'ALM_harmonised': 6114 derived, 0 missing
. Range 9061.706 - 27682.402 |
  Mean ± SD: 16501.66 ± 2614.973
. 'ALM_HT2_harmonised': 6114 derived, 0 missing
. Range 3.811 - 10.943 |
  Mean ± SD: 6.377 ± 0.871
. 'ALM_BMI_harmonised': 6114 derived, 0 missing
. Range 0.354 - 1.054 |
  Mean ± SD: 0.646 ± 0.101
. 'ALM_WT_harmonised': 6114 derived, 0 missing
. Range 0.165 - 0.381 |
  Mean ± SD: 0.249 ± 0.028
. Sanity check passed: ALM_harmonised == ALM for all 2824 Hologic rows.

```

```

. Agreement 'ALM' vs 'ALM_source' (n = 5636):
. < 0.001: 5629 (99.9%)
. 0.001 to 0.1: 0 (0%)
. >= 0.1: 7 (0.1%)
. Median |diff|: 0 | Max: 343.6143
7 row(s) differ by >= 0.1 between derived and source.
. Agreement 'ALM_HT2' vs 'ALM_HT2_source' (n = 5636):
. < 0.001: 5629 (99.9%)
. 0.001 to 0.1: 6 (0.1%)
. >= 0.1: 1 (0%)
. Median |diff|: 0 | Max: 0.1385
1 row(s) differ by >= 0.1 between derived and source.
. Agreement 'ALM_BMI' vs 'ALM_BMI_source' (n = 5635):
. < 0.001: 5629 (99.9%)
. 0.001 to 0.1: 6 (0.1%)
. >= 0.1: 0 (0%)
. Median |diff|: 1e-04 | Max: 0.0149
. Agreement 'ALM_WT' vs 'ALM_WT_source' (n = 5636):
. < 0.001: 5635 (100%)
. 0.001 to 0.1: 1 (0%)
. >= 0.1: 0 (0%)
. Median |diff|: 0 | Max: 0.0061

-> 63 targets ..... [14.9s, 2+, 15-]

.. Split ALM by DXA method ..
-> 63 targets ..... [14.9s, 2+, 15-]

-> 63 targets ..... [14.9s, 2+, 15-]
. Method-specific ALM columns created:
. Hologic and Lunar splits for ALM, ALM_HT2, ALM_BMI, ALM_WT

-> 63 targets ..... [14.9s, 2+, 15-]

.. Derive Age ..
-> 63 targets ..... [14.9s, 2+, 15-]

-> 63 targets ..... [14.9s, 2+, 15-]
. Age derived from datbirth (Baseline, filled forward/back) and exam_date_iso.
. Total rows: 6114 | derived: 6114 | missing: 0
. Age range (non-missing): 50.2 - 91.5 years
. Mean ± SD: 68.6 ± 7.9 years
. Agreement with Age_source (rows with both non-missing, n = 6114):
. Difference < 0.1 yr (effectively identical): 6066 (99.2%)
. Difference 0.1-1 yr (minor rounding/timing): 47 (0.8%)
. Difference >= 1 yr (meaningful mismatch) : 1 (0%)
. Median absolute difference: 0.028 years
. Max absolute difference : 4.328 years
[4/20] deriving ...

-> 63 targets ..... [14.9s, 2+, 15-]

.. Derive BMI ..

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```

-> 63 targets ..... [14.9s, 2+, 15-]

-> 63 targets ..... [14.9s, 2+, 15-]
. BMI derived from Weight and Height.
. Total rows: 6114 | derived: 6114 | missing: 0
. BMI range (non-missing): 15.1 - 45.5 kg/m²
. Mean ± SD: 25.9 ± 4.7 kg/m²
. Agreement with BMI_source (rows with both non-missing, n = 6112):
. Difference < 0.01 (effectively identical): 6099 (99.8%)
. Difference 0.01-0.5 (minor rounding): 13 (0.2%)
. Difference >= 0.5 (meaningful mismatch): 0 (0%)
. Median absolute difference: 0.002 kg/m²
. Max absolute difference : 0.17 kg/m²

-> 63 targets ..... [14.9s, 2+, 15-]

.. Deriving BMI category ..
-> 63 targets ..... [14.9s, 2+, 15-]

-> 63 targets ..... [14.9s, 2+, 15-]
. derive_bmi_category: BMI categories derived.
. Processed 6114 rows (0 missing BMI).
  Prevalence of Obesity: 18.5%
  Prevalence of Overweight + Obesity: 53.4%

-> 63 targets ..... [14.9s, 2+, 15-]

.. Derive ALM Indices ..
-> 63 targets ..... [14.9s, 2+, 15-]

-> 63 targets ..... [14.9s, 2+, 15-]

.. Raw indices (native DXA platform)
-> 63 targets ..... [14.9s, 2+, 15-]
. 'ALM': 6114 derived, 0 missing
. Range 9061.706 - 27682.402 |
  Mean ± SD: 17002.239 ± 2587.838
. 'ALM_HT2': 6114 derived, 0 missing
. Range 4.1 - 10.943 |
  Mean ± SD: 6.572 ± 0.858
. 'ALM_BMI': 6114 derived, 0 missing
. Range 0.379 - 1.054 |
  Mean ± SD: 0.666 ± 0.101
. 'ALM_WT': 6114 derived, 0 missing
. Range 0.178 - 0.381 |
  Mean ± SD: 0.257 ± 0.028

-> 63 targets ..... [14.9s, 2+, 15-]

.. Harmonised indices (Lunar converted to Hologic scale)
-> 63 targets ..... [14.9s, 2+, 15-]
. 'ALM_harmonised': 6114 derived, 0 missing

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```

. Range 9061.706 - 27682.402 |
  Mean ± SD: 16504.45 ± 2615.828
. 'ALM_HT2_harmonised': 6114 derived, 0 missing
. Range 3.811 - 10.943 |
  Mean ± SD: 6.378 ± 0.871
. 'ALM_BMI_harmonised': 6114 derived, 0 missing
. Range 0.354 - 1.054 |
  Mean ± SD: 0.646 ± 0.101
. 'ALM_WT_harmonised': 6114 derived, 0 missing
. Range 0.168 - 0.381 |
  Mean ± SD: 0.249 ± 0.028
. Sanity check passed: ALM_harmonised == ALM for all 2824 Hologic rows.
. Agreement 'ALM' vs 'ALM_source' (n = 5636):
. < 0.001: 5629 (99.9%)
. 0.001 to 0.1: 0 (0%)
. >= 0.1: 7 (0.1%)
. Median |diff|: 0 | Max: 387.8053
7 row(s) differ by >= 0.1 between derived and source.
. Agreement 'ALM_HT2' vs 'ALM_HT2_source' (n = 5636):
. < 0.001: 5629 (99.9%)
. 0.001 to 0.1: 6 (0.1%)
. >= 0.1: 1 (0%)
. Median |diff|: 0 | Max: 0.1563
1 row(s) differ by >= 0.1 between derived and source.
. Agreement 'ALM_BMI' vs 'ALM_BMI_source' (n = 5635):
. < 0.001: 5629 (99.9%)
. 0.001 to 0.1: 6 (0.1%)
. >= 0.1: 0 (0%)
. Median |diff|: 1e-04 | Max: 0.0171
. Agreement 'ALM_WT' vs 'ALM_WT_source' (n = 5636):
. < 0.001: 5635 (100%)
. 0.001 to 0.1: 1 (0%)
. >= 0.1: 0 (0%)
. Median |diff|: 0 | Max: 0.0069

-> 63 targets ..... [14.9s, 2+, 15-]

.. Split ALM by DXA method ..
-> 63 targets ..... [14.9s, 2+, 15-]

-> 63 targets ..... [14.9s, 2+, 15-]
. Method-specific ALM columns created:
. Hologic and Lunar splits for ALM, ALM_HT2, ALM_BMI, ALM_WT

-> 63 targets ..... [14.9s, 2+, 15-]

.. Derive Age ..
-> 63 targets ..... [14.9s, 2+, 15-]

-> 63 targets ..... [14.9s, 2+, 15-]
. Age derived from datbirth (Baseline, filled forward/back) and exam_date_iso.
. Total rows: 6114 | derived: 6114 | missing: 0
. Age range (non-missing): 50.2 - 91.5 years
. Mean ± SD: 68.6 ± 7.9 years

```

```

. Agreement with Age_source (rows with both non-missing, n = 6114):
. Difference < 0.1 yr (effectively identical): 6066 (99.2%)
. Difference 0.1-1 yr (minor rounding/timing): 47 (0.8%)
. Difference >= 1 yr (meaningful mismatch) : 1 (0%)
. Median absolute difference: 0.028 years
. Max absolute difference : 4.328 years
[5/20] deriving ...

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```

-> 63 targets ..... [14.9s, 2+, 15-]

```

```

.. Derive BMI ..

```

```

-> 63 targets ..... [14.9s, 2+, 15-]

```

```

-> 63 targets ..... [14.9s, 2+, 15-]
. BMI derived from Weight and Height.
. Total rows: 6114 | derived: 6114 | missing: 0
. BMI range (non-missing): 15.1 - 45.5 kg/m²
. Mean ± SD: 25.9 ± 4.7 kg/m²
. Agreement with BMI_source (rows with both non-missing, n = 6112):
. Difference < 0.01 (effectively identical): 6099 (99.8%)
. Difference 0.01-0.5 (minor rounding): 13 (0.2%)
. Difference >= 0.5 (meaningful mismatch): 0 (0%)
. Median absolute difference: 0.002 kg/m²
. Max absolute difference : 0.17 kg/m²

```

```

-> 63 targets ..... [14.9s, 2+, 15-]

```

```

.. Deriving BMI category ..

```

```

-> 63 targets ..... [14.9s, 2+, 15-]

```

```

-> 63 targets ..... [14.9s, 2+, 15-]
. derive_bmi_category: BMI categories derived.
. Processed 6114 rows (0 missing BMI).
  Prevalence of Obesity: 18.5%
  Prevalence of Overweight + Obesity: 53.4%

```

```

-> 63 targets ..... [14.9s, 2+, 15-]

```

```

.. Derive ALM Indices ..

```

```

-> 63 targets ..... [14.9s, 2+, 15-]

```

```

-> 63 targets ..... [14.9s, 2+, 15-]

```

```

.. Raw indices (native DXA platform)

```

```

-> 63 targets ..... [14.9s, 2+, 15-]

```

```

. 'ALM': 6114 derived, 0 missing
. Range 9061.706 - 27682.402 |
  Mean ± SD: 17012.691 ± 2584.377
. 'ALM_HT2': 6114 derived, 0 missing
. Range 4.1 - 10.943 |
  Mean ± SD: 6.576 ± 0.856
. 'ALM_BMI': 6114 derived, 0 missing

```

```

. Range 0.379 - 1.054 |
  Mean ± SD: 0.667 ± 0.101
. 'ALM_WT': 6114 derived, 0 missing
. Range 0.178 - 0.381 |
  Mean ± SD: 0.257 ± 0.028

-> 63 targets ..... [14.9s, 2+, 15-]

.. Harmonised indices (Lunar converted to Hologic scale)
-> 63 targets ..... [14.9s, 2+, 15-]
. 'ALM_harmonised': 6114 derived, 0 missing
. Range 9061.706 - 27682.402 |
  Mean ± SD: 16514.892 ± 2614.316
. 'ALM_HT2_harmonised': 6114 derived, 0 missing
. Range 3.811 - 10.943 |
  Mean ± SD: 6.382 ± 0.869
. 'ALM_BMI_harmonised': 6114 derived, 0 missing
. Range 0.354 - 1.054 |
  Mean ± SD: 0.647 ± 0.101
. 'ALM_WT_harmonised': 6114 derived, 0 missing
. Range 0.168 - 0.381 |
  Mean ± SD: 0.249 ± 0.028
. Sanity check passed: ALM_harmonised == ALM for all 2824 Hologic rows.
. Agreement 'ALM' vs 'ALM_source' (n = 5636):
. < 0.001: 5629 (99.9%)
. 0.001 to 0.1: 0 (0%)
. >= 0.1: 7 (0.1%)
. Median |diff|: 0 | Max: 91.5323
7 row(s) differ by >= 0.1 between derived and source.
. Agreement 'ALM_HT2' vs 'ALM_HT2_source' (n = 5636):
. < 0.001: 5629 (99.9%)
. 0.001 to 0.1: 7 (0.1%)
. >= 0.1: 0 (0%)
. Median |diff|: 0 | Max: 0.0369
. Agreement 'ALM_BMI' vs 'ALM_BMI_source' (n = 5635):
. < 0.001: 5629 (99.9%)
. 0.001 to 0.1: 6 (0.1%)
. >= 0.1: 0 (0%)
. Median |diff|: 1e-04 | Max: 0.0052
. Agreement 'ALM_WT' vs 'ALM_WT_source' (n = 5636):
. < 0.001: 5635 (100%)
. 0.001 to 0.1: 1 (0%)
. >= 0.1: 0 (0%)
. Median |diff|: 0 | Max: 0.0016

-> 63 targets ..... [14.9s, 2+, 15-]

.. Split ALM by DXA method ..
-> 63 targets ..... [14.9s, 2+, 15-]

-> 63 targets ..... [14.9s, 2+, 15-]
. Method-specific ALM columns created:
. Hologic and Lunar splits for ALM, ALM_HT2, ALM_BMI, ALM_WT

```

```

-> 63 targets ..... [14.9s, 2+, 15-]

.. Derive Age ..
-> 63 targets ..... [14.9s, 2+, 15-]

-> 63 targets ..... [14.9s, 2+, 15-]
. Age derived from datbirth (Baseline, filled forward/back) and exam_date_iso.
. Total rows: 6114 | derived: 6114 | missing: 0
. Age range (non-missing): 50.2 - 91.5 years
. Mean  $\pm$  SD: 68.6  $\pm$  7.9 years
. Agreement with Age_source (rows with both non-missing, n = 6114):
. Difference < 0.1 yr (effectively identical): 6066 (99.2%)
. Difference 0.1-1 yr (minor rounding/timing): 47 (0.8%)
. Difference  $\geq$  1 yr (meaningful mismatch) : 1 (0%)
. Median absolute difference: 0.028 years
. Max absolute difference : 4.328 years
[6/20] deriving ...

-> 63 targets ..... [14.9s, 2+, 15-]

.. Derive BMI ..
-> 63 targets ..... [14.9s, 2+, 15-]

-> 63 targets ..... [14.9s, 2+, 15-]
. BMI derived from Weight and Height.
. Total rows: 6114 | derived: 6114 | missing: 0
. BMI range (non-missing): 15.1 - 45.5 kg/m2
. Mean  $\pm$  SD: 25.9  $\pm$  4.7 kg/m2
. Agreement with BMI_source (rows with both non-missing, n = 6112):
. Difference < 0.01 (effectively identical): 6099 (99.8%)
. Difference 0.01-0.5 (minor rounding): 13 (0.2%)
. Difference  $\geq$  0.5 (meaningful mismatch): 0 (0%)
. Median absolute difference: 0.002 kg/m2
. Max absolute difference : 0.17 kg/m2

-> 63 targets ..... [14.9s, 2+, 15-]

.. Deriving BMI category ..
-> 63 targets ..... [14.9s, 2+, 15-]

-> 63 targets ..... [14.9s, 2+, 15-]
. derive_bmi_category: BMI categories derived.
. Processed 6114 rows (0 missing BMI).
  Prevalence of Obesity: 18.5%
  Prevalence of Overweight + Obesity: 53.4%

-> 63 targets ..... [14.9s, 2+, 15-]

.. Derive ALM Indices ..
-> 63 targets ..... [14.9s, 2+, 15-]

```

```

-> 63 targets ..... [14.9s, 2+, 15-]

.. Raw indices (native DXA platform)
-> 63 targets ..... [14.9s, 2+, 15-]
. 'ALM': 6114 derived, 0 missing
. Range 9061.706 - 27682.402 |
  Mean ± SD: 17005.045 ± 2591.797
. 'ALM_HT2': 6114 derived, 0 missing
. Range 4.1 - 10.943 |
  Mean ± SD: 6.573 ± 0.861
. 'ALM_BMI': 6114 derived, 0 missing
. Range 0.379 - 1.054 |
  Mean ± SD: 0.666 ± 0.101
. 'ALM_WT': 6114 derived, 0 missing
. Range 0.178 - 0.381 |
  Mean ± SD: 0.257 ± 0.028

-> 63 targets ..... [14.9s, 2+, 15-]

.. Harmonised indices (Lunar converted to Hologic scale)
-> 63 targets ..... [14.9s, 2+, 15-]
. 'ALM_harmonised': 6114 derived, 0 missing
. Range 9061.706 - 27682.402 |
  Mean ± SD: 16507.253 ± 2620.274
. 'ALM_HT2_harmonised': 6114 derived, 0 missing
. Range 3.811 - 10.943 |
  Mean ± SD: 6.379 ± 0.873
. 'ALM_BMI_harmonised': 6114 derived, 0 missing
. Range 0.354 - 1.054 |
  Mean ± SD: 0.647 ± 0.101
. 'ALM_WT_harmonised': 6114 derived, 0 missing
. Range 0.168 - 0.381 |
  Mean ± SD: 0.249 ± 0.028
. Sanity check passed: ALM_harmonised == ALM for all 2824 Hologic rows.
. Agreement 'ALM' vs 'ALM_source' (n = 5636):
. < 0.001: 5629 (99.9%)
. 0.001 to 0.1: 0 (0%)
. >= 0.1: 7 (0.1%)
. Median |diff|: 0 | Max: 54.9565
7 row(s) differ by >= 0.1 between derived and source.
. Agreement 'ALM_HT2' vs 'ALM_HT2_source' (n = 5636):
. < 0.001: 5629 (99.9%)
. 0.001 to 0.1: 7 (0.1%)
. >= 0.1: 0 (0%)
. Median |diff|: 0 | Max: 0.0222
. Agreement 'ALM_BMI' vs 'ALM_BMI_source' (n = 5635):
. < 0.001: 5629 (99.9%)
. 0.001 to 0.1: 6 (0.1%)
. >= 0.1: 0 (0%)
. Median |diff|: 1e-04 | Max: 0.0052
. Agreement 'ALM_WT' vs 'ALM_WT_source' (n = 5636):
. < 0.001: 5636 (100%)
. 0.001 to 0.1: 0 (0%)
. >= 0.1: 0 (0%)
. Median |diff|: 0 | Max: 0.001

```

```

-> 63 targets ..... [14.9s, 2+, 15-]

.. Split ALM by DXA method ..
-> 63 targets ..... [14.9s, 2+, 15-]

-> 63 targets ..... [14.9s, 2+, 15-]
. Method-specific ALM columns created:
. Hologic and Lunar splits for ALM, ALM_HT2, ALM_BMI, ALM_WT

-> 63 targets ..... [14.9s, 2+, 15-]

.. Derive Age ..
-> 63 targets ..... [14.9s, 2+, 15-]

-> 63 targets ..... [14.9s, 2+, 15-]
. Age derived from datbirth (Baseline, filled forward/back) and exam_date_iso.
. Total rows: 6114 | derived: 6114 | missing: 0
. Age range (non-missing): 50.2 - 91.5 years
. Mean ± SD: 68.6 ± 7.9 years
. Agreement with Age_source (rows with both non-missing, n = 6114):
. Difference < 0.1 yr (effectively identical): 6066 (99.2%)
. Difference 0.1-1 yr (minor rounding/timing): 47 (0.8%)
. Difference >= 1 yr (meaningful mismatch) : 1 (0%)
. Median absolute difference: 0.028 years
. Max absolute difference : 4.328 years
[7/20] deriving ...

-> 63 targets ..... [14.9s, 2+, 15-]

.. Derive BMI ..
-> 63 targets ..... [14.9s, 2+, 15-]

-> 63 targets ..... [14.9s, 2+, 15-]
. BMI derived from Weight and Height.
. Total rows: 6114 | derived: 6114 | missing: 0
. BMI range (non-missing): 15.1 - 45.5 kg/m²
. Mean ± SD: 25.9 ± 4.7 kg/m²
. Agreement with BMI_source (rows with both non-missing, n = 6112):
. Difference < 0.01 (effectively identical): 6099 (99.8%)
. Difference 0.01-0.5 (minor rounding): 13 (0.2%)
. Difference >= 0.5 (meaningful mismatch): 0 (0%)
. Median absolute difference: 0.002 kg/m²
. Max absolute difference : 0.17 kg/m²

-> 63 targets ..... [14.9s, 2+, 15-]

.. Deriving BMI category ..
-> 63 targets ..... [14.9s, 2+, 15-]

-> 63 targets ..... [14.9s, 2+, 15-]
. derive_bmi_category: BMI categories derived.

```

```

. Processed 6114 rows (0 missing BMI).
  Prevalence of Obesity: 18.5%
  Prevalence of Overweight + Obesity: 53.4%

-> 63 targets ..... [14.9s, 2+, 15-]

.. Derive ALM Indices ..
-> 63 targets ..... [14.9s, 2+, 15-]

-> 63 targets ..... [14.9s, 2+, 15-]

.. Raw indices (native DXA platform)
-> 63 targets ..... [14.9s, 2+, 15-]
. 'ALM': 6114 derived, 0 missing
. Range 9061.706 - 27682.402 |
  Mean ± SD: 17008.904 ± 2591.076
. 'ALM_HT2': 6114 derived, 0 missing
. Range 4.1 - 10.943 |
  Mean ± SD: 6.574 ± 0.858
. 'ALM_BMI': 6114 derived, 0 missing
. Range 0.379 - 1.054 |
  Mean ± SD: 0.666 ± 0.101
. 'ALM_WT': 6114 derived, 0 missing
. Range 0.178 - 0.381 |
  Mean ± SD: 0.257 ± 0.028

-> 63 targets ..... [14.9s, 2+, 15-]

.. Harmonised indices (Lunar converted to Hologic scale)
-> 63 targets ..... [14.9s, 2+, 15-]
. 'ALM_harmonised': 6114 derived, 0 missing
. Range 9061.706 - 27682.402 |
  Mean ± SD: 16511.103 ± 2620.196
. 'ALM_HT2_harmonised': 6114 derived, 0 missing
. Range 3.811 - 10.943 |
  Mean ± SD: 6.381 ± 0.872
. 'ALM_BMI_harmonised': 6114 derived, 0 missing
. Range 0.354 - 1.054 |
  Mean ± SD: 0.647 ± 0.101
. 'ALM_WT_harmonised': 6114 derived, 0 missing
. Range 0.168 - 0.381 |
  Mean ± SD: 0.249 ± 0.028
. Sanity check passed: ALM_harmonised == ALM for all 2824 Hologic rows.
. Agreement 'ALM' vs 'ALM_source' (n = 5636):
. < 0.001: 5629 (99.9%)
. 0.001 to 0.1: 0 (0%)
. >= 0.1: 7 (0.1%)
. Median |diff|: 0 | Max: 162.1173
7 row(s) differ by >= 0.1 between derived and source.
. Agreement 'ALM_HT2' vs 'ALM_HT2_source' (n = 5636):
. < 0.001: 5629 (99.9%)
. 0.001 to 0.1: 7 (0.1%)
. >= 0.1: 0 (0%)
. Median |diff|: 0 | Max: 0.0653
. Agreement 'ALM_BMI' vs 'ALM_BMI_source' (n = 5635):

```

```

. < 0.001: 5629 (99.9%)
. 0.001 to 0.1: 6 (0.1%)
. >= 0.1: 0 (0%)
. Median |diff|: 1e-04 | Max: 0.007
. Agreement 'ALM_WT' vs 'ALM_WT_source' (n = 5636):
. < 0.001: 5635 (100%)
. 0.001 to 0.1: 1 (0%)
. >= 0.1: 0 (0%)
. Median |diff|: 0 | Max: 0.0029

-> 63 targets ..... [14.9s, 2+, 15-]

.. Split ALM by DXA method ..
-> 63 targets ..... [14.9s, 2+, 15-]

-> 63 targets ..... [14.9s, 2+, 15-]
. Method-specific ALM columns created:
. Hologic and Lunar splits for ALM, ALM_HT2, ALM_BMI, ALM_WT

-> 63 targets ..... [14.9s, 2+, 15-]

.. Derive Age ..
-> 63 targets ..... [14.9s, 2+, 15-]

-> 63 targets ..... [14.9s, 2+, 15-]
. Age derived from datbirth (Baseline, filled forward/back) and exam_date_iso.
. Total rows: 6114 | derived: 6114 | missing: 0
. Age range (non-missing): 50.2 - 91.5 years
. Mean ± SD: 68.6 ± 7.9 years
. Agreement with Age_source (rows with both non-missing, n = 6114):
. Difference < 0.1 yr (effectively identical): 6066 (99.2%)
. Difference 0.1-1 yr (minor rounding/timing): 47 (0.8%)
. Difference >= 1 yr (meaningful mismatch) : 1 (0%)
. Median absolute difference: 0.028 years
. Max absolute difference : 4.328 years
[8/20] deriving ...

-> 63 targets ..... [14.9s, 2+, 15-]

.. Derive BMI ..
-> 63 targets ..... [14.9s, 2+, 15-]

-> 63 targets ..... [14.9s, 2+, 15-]
. BMI derived from Weight and Height.
. Total rows: 6114 | derived: 6114 | missing: 0
. BMI range (non-missing): 15.1 - 45.5 kg/m²
. Mean ± SD: 25.9 ± 4.7 kg/m²
. Agreement with BMI_source (rows with both non-missing, n = 6112):
. Difference < 0.01 (effectively identical): 6099 (99.8%)
. Difference 0.01-0.5 (minor rounding): 13 (0.2%)
. Difference >= 0.5 (meaningful mismatch): 0 (0%)
. Median absolute difference: 0.002 kg/m²

```

```

. Max absolute difference : 0.17 kg/m²

-> 63 targets ..... [14.9s, 2+, 15-]

.. Deriving BMI category ..
-> 63 targets ..... [14.9s, 2+, 15-]

-> 63 targets ..... [14.9s, 2+, 15-]
. derive_bmi_category: BMI categories derived.
. Processed 6114 rows (0 missing BMI).
  Prevalence of Obesity: 18.5%
  Prevalence of Overweight + Obesity: 53.4%

-> 63 targets ..... [14.9s, 2+, 15-]

.. Derive ALM Indices ..
-> 63 targets ..... [14.9s, 2+, 15-]

-> 63 targets ..... [14.9s, 2+, 15-]

.. Raw indices (native DXA platform)
-> 63 targets ..... [14.9s, 2+, 15-]
. 'ALM': 6114 derived, 0 missing
. Range 9061.706 - 27682.402 |
  Mean ± SD: 17008.758 ± 2590.714
. 'ALM_HT2': 6114 derived, 0 missing
. Range 4.1 - 10.943 |
  Mean ± SD: 6.574 ± 0.859
. 'ALM_BMI': 6114 derived, 0 missing
. Range 0.379 - 1.054 |
  Mean ± SD: 0.666 ± 0.101
. 'ALM_WT': 6114 derived, 0 missing
. Range 0.178 - 0.381 |
  Mean ± SD: 0.257 ± 0.028

-> 63 targets ..... [14.9s, 2+, 15-]

.. Harmonised indices (Lunar converted to Hologic scale)
-> 63 targets ..... [14.9s, 2+, 15-]
. 'ALM_harmonised': 6114 derived, 0 missing
. Range 9061.706 - 27682.402 |
  Mean ± SD: 16510.958 ± 2619.808
. 'ALM_HT2_harmonised': 6114 derived, 0 missing
. Range 3.811 - 10.943 |
  Mean ± SD: 6.381 ± 0.872
. 'ALM_BMI_harmonised': 6114 derived, 0 missing
. Range 0.354 - 1.054 |
  Mean ± SD: 0.647 ± 0.101
. 'ALM_WT_harmonised': 6114 derived, 0 missing
. Range 0.168 - 0.381 |
  Mean ± SD: 0.249 ± 0.028
. Sanity check passed: ALM_harmonised == ALM for all 2824 Hologic rows.
. Agreement 'ALM' vs 'ALM_source' (n = 5636):

```

```

. < 0.001: 5629 (99.9%)
. 0.001 to 0.1: 0 (0%)
. >= 0.1: 7 (0.1%)
. Median |diff|: 0 | Max: 357.0801
7 row(s) differ by >= 0.1 between derived and source.
. Agreement 'ALM_HT2' vs 'ALM_HT2_source' (n = 5636):
. < 0.001: 5629 (99.9%)
. 0.001 to 0.1: 6 (0.1%)
. >= 0.1: 1 (0%)
. Median |diff|: 0 | Max: 0.1439
1 row(s) differ by >= 0.1 between derived and source.
. Agreement 'ALM_BMI' vs 'ALM_BMI_source' (n = 5635):
. < 0.001: 5629 (99.9%)
. 0.001 to 0.1: 6 (0.1%)
. >= 0.1: 0 (0%)
. Median |diff|: 1e-04 | Max: 0.0155
. Agreement 'ALM_WT' vs 'ALM_WT_source' (n = 5636):
. < 0.001: 5635 (100%)
. 0.001 to 0.1: 1 (0%)
. >= 0.1: 0 (0%)
. Median |diff|: 0 | Max: 0.0063

```

```

-> 63 targets ..... [14.9s, 2+, 15-]

```

```

.. Split ALM by DXA method ..
-> 63 targets ..... [14.9s, 2+, 15-]

```

```

-> 63 targets ..... [14.9s, 2+, 15-]
. Method-specific ALM columns created:
. Hologic and Lunar splits for ALM, ALM_HT2, ALM_BMI, ALM_WT

```

```

-> 63 targets ..... [14.9s, 2+, 15-]

```

```

.. Derive Age ..
-> 63 targets ..... [14.9s, 2+, 15-]

```

```

-> 63 targets ..... [14.9s, 2+, 15-]
. Age derived from datbirth (Baseline, filled forward/back) and exam_date_iso.
. Total rows: 6114 | derived: 6114 | missing: 0
. Age range (non-missing): 50.2 - 91.5 years
. Mean ± SD: 68.6 ± 7.9 years
. Agreement with Age_source (rows with both non-missing, n = 6114):
. Difference < 0.1 yr (effectively identical): 6066 (99.2%)
. Difference 0.1-1 yr (minor rounding/timing): 47 (0.8%)
. Difference >= 1 yr (meaningful mismatch) : 1 (0%)
. Median absolute difference: 0.028 years
. Max absolute difference : 4.328 years
[9/20] deriving ...

```

```

-> 63 targets ..... [14.9s, 2+, 15-]

```

```

.. Derive BMI ..
-> 63 targets ..... [14.9s, 2+, 15-]

```

```

-> 63 targets ..... [14.9s, 2+, 15-]
. BMI derived from Weight and Height.
. Total rows: 6114 | derived: 6114 | missing: 0
. BMI range (non-missing): 15.1 - 45.5 kg/m²
. Mean ± SD: 25.9 ± 4.7 kg/m²
. Agreement with BMI_source (rows with both non-missing, n = 6112):
. Difference < 0.01 (effectively identical): 6099 (99.8%)
. Difference 0.01-0.5 (minor rounding): 13 (0.2%)
. Difference >= 0.5 (meaningful mismatch): 0 (0%)
. Median absolute difference: 0.002 kg/m²
. Max absolute difference : 0.17 kg/m²

```

```

-> 63 targets ..... [14.9s, 2+, 15-]

```

```

.. Deriving BMI category ..

```

```

-> 63 targets ..... [14.9s, 2+, 15-]

```

```

-> 63 targets ..... [14.9s, 2+, 15-]

```

```

. derive_bmi_category: BMI categories derived.
. Processed 6114 rows (0 missing BMI).
  Prevalence of Obesity: 18.5%
  Prevalence of Overweight + Obesity: 53.4%

```

```

-> 63 targets ..... [14.9s, 2+, 15-]

```

```

.. Derive ALM Indices ..

```

```

-> 63 targets ..... [14.9s, 2+, 15-]

```

```

-> 63 targets ..... [14.9s, 2+, 15-]

```

```

.. Raw indices (native DXA platform)

```

```

-> 63 targets ..... [14.9s, 2+, 15-]

```

```

. 'ALM': 6114 derived, 0 missing
. Range 9061.706 - 27682.402 |
  Mean ± SD: 17014.112 ± 2586.99
. 'ALM_HT2': 6114 derived, 0 missing
. Range 4.1 - 10.943 |
  Mean ± SD: 6.576 ± 0.859
. 'ALM_BMI': 6114 derived, 0 missing
. Range 0.379 - 1.054 |
  Mean ± SD: 0.667 ± 0.101
. 'ALM_WT': 6114 derived, 0 missing
. Range 0.178 - 0.381 |
  Mean ± SD: 0.257 ± 0.028

```

```

-> 63 targets ..... [14.9s, 2+, 15-]

```

```

.. Harmonised indices (Lunar converted to Hologic scale)

```

```

-> 63 targets ..... [14.9s, 2+, 15-]

```

```

. 'ALM_harmonised': 6114 derived, 0 missing
. Range 9061.706 - 27682.402 |

```

```

Mean ± SD: 16516.316 ± 2617.196
. 'ALM_HT2_harmonised': 6114 derived, 0 missing
. Range 3.811 - 10.943 |
  Mean ± SD: 6.383 ± 0.872
. 'ALM_BMI_harmonised': 6114 derived, 0 missing
. Range 0.354 - 1.054 |
  Mean ± SD: 0.647 ± 0.101
. 'ALM_WT_harmonised': 6114 derived, 0 missing
. Range 0.168 - 0.381 |
  Mean ± SD: 0.249 ± 0.028
. Sanity check passed: ALM_harmonised == ALM for all 2824 Hologic rows.
. Agreement 'ALM' vs 'ALM_source' (n = 5636):
. < 0.001: 5629 (99.9%)
. 0.001 to 0.1: 0 (0%)
. >= 0.1: 7 (0.1%)
. Median |diff|: 0 | Max: 138.4331
7 row(s) differ by >= 0.1 between derived and source.
. Agreement 'ALM_HT2' vs 'ALM_HT2_source' (n = 5636):
. < 0.001: 5629 (99.9%)
. 0.001 to 0.1: 7 (0.1%)
. >= 0.1: 0 (0%)
. Median |diff|: 0 | Max: 0.0558
. Agreement 'ALM_BMI' vs 'ALM_BMI_source' (n = 5635):
. < 0.001: 5629 (99.9%)
. 0.001 to 0.1: 6 (0.1%)
. >= 0.1: 0 (0%)
. Median |diff|: 1e-04 | Max: 0.0062
. Agreement 'ALM_WT' vs 'ALM_WT_source' (n = 5636):
. < 0.001: 5635 (100%)
. 0.001 to 0.1: 1 (0%)
. >= 0.1: 0 (0%)
. Median |diff|: 0 | Max: 0.0025

-> 63 targets ..... [14.9s, 2+, 15-]

.. Split ALM by DXA method ..
-> 63 targets ..... [14.9s, 2+, 15-]

-> 63 targets ..... [14.9s, 2+, 15-]
. Method-specific ALM columns created:
. Hologic and Lunar splits for ALM, ALM_HT2, ALM_BMI, ALM_WT

-> 63 targets ..... [14.9s, 2+, 15-]

.. Derive Age ..
-> 63 targets ..... [14.9s, 2+, 15-]

-> 63 targets ..... [14.9s, 2+, 15-]
. Age derived from datbirth (Baseline, filled forward/back) and exam_date_iso.
. Total rows: 6114 | derived: 6114 | missing: 0
. Age range (non-missing): 50.2 - 91.5 years
. Mean ± SD: 68.6 ± 7.9 years
. Agreement with Age_source (rows with both non-missing, n = 6114):
. Difference < 0.1 yr (effectively identical): 6066 (99.2%)

```

```

. Difference 0.1-1 yr (minor rounding/timing): 47 (0.8%)
. Difference >= 1 yr (meaningful mismatch) : 1 (0%)
. Median absolute difference: 0.028 years
. Max absolute difference : 4.328 years
[10/20] deriving ...

-> 63 targets ..... [14.9s, 2+, 15-]

.. Derive BMI ..
-> 63 targets ..... [14.9s, 2+, 15-]

-> 63 targets ..... [14.9s, 2+, 15-]
. BMI derived from Weight and Height.
. Total rows: 6114 | derived: 6114 | missing: 0
. BMI range (non-missing): 15.1 - 45.5 kg/m²
. Mean ± SD: 25.9 ± 4.7 kg/m²
. Agreement with BMI_source (rows with both non-missing, n = 6112):
. Difference < 0.01 (effectively identical): 6099 (99.8%)
. Difference 0.01-0.5 (minor rounding): 13 (0.2%)
. Difference >= 0.5 (meaningful mismatch): 0 (0%)
. Median absolute difference: 0.002 kg/m²
. Max absolute difference : 0.17 kg/m²

-> 63 targets ..... [14.9s, 2+, 15-]

.. Deriving BMI category ..
-> 63 targets ..... [14.9s, 2+, 15-]

-> 63 targets ..... [14.9s, 2+, 15-]
. derive_bmi_category: BMI categories derived.
. Processed 6114 rows (0 missing BMI).
  Prevalence of Obesity: 18.5%
  Prevalence of Overweight + Obesity: 53.4%

-> 63 targets ..... [14.9s, 2+, 15-]

.. Derive ALM Indices ..
-> 63 targets ..... [14.9s, 2+, 15-]

-> 63 targets ..... [14.9s, 2+, 15-]

.. Raw indices (native DXA platform)
-> 63 targets ..... [14.9s, 2+, 15-]
. 'ALM': 6114 derived, 0 missing
. Range 9061.706 - 27682.402 |
  Mean ± SD: 17002.904 ± 2590.038
. 'ALM_HT2': 6114 derived, 0 missing
. Range 4.1 - 10.943 |
  Mean ± SD: 6.572 ± 0.858
. 'ALM_BMI': 6114 derived, 0 missing
. Range 0.379 - 1.054 |
  Mean ± SD: 0.666 ± 0.101

```

```

. 'ALM_WT': 6114 derived, 0 missing
. Range 0.178 - 0.381 |
  Mean ± SD: 0.257 ± 0.028

-> 63 targets ..... [14.9s, 2+, 15-]

.. Harmonised indices (Lunar converted to Hologic scale)
-> 63 targets ..... [14.9s, 2+, 15-]
. 'ALM_harmonised': 6114 derived, 0 missing
. Range 9061.706 - 27682.402 |
  Mean ± SD: 16505.118 ± 2618.2
. 'ALM_HT2_harmonised': 6114 derived, 0 missing
. Range 3.811 - 10.943 |
  Mean ± SD: 6.378 ± 0.871
. 'ALM_BMI_harmonised': 6114 derived, 0 missing
. Range 0.354 - 1.054 |
  Mean ± SD: 0.646 ± 0.101
. 'ALM_WT_harmonised': 6114 derived, 0 missing
. Range 0.168 - 0.381 |
  Mean ± SD: 0.249 ± 0.028
. Sanity check passed: ALM_harmonised == ALM for all 2824 Hologic rows.
. Agreement 'ALM' vs 'ALM_source' (n = 5636):
. < 0.001: 5629 (99.9%)
. 0.001 to 0.1: 0 (0%)
. >= 0.1: 7 (0.1%)
. Median |diff|: 0 | Max: 132.6214
7 row(s) differ by >= 0.1 between derived and source.
. Agreement 'ALM_HT2' vs 'ALM_HT2_source' (n = 5636):
. < 0.001: 5629 (99.9%)
. 0.001 to 0.1: 7 (0.1%)
. >= 0.1: 0 (0%)
. Median |diff|: 0 | Max: 0.0535
. Agreement 'ALM_BMI' vs 'ALM_BMI_source' (n = 5635):
. < 0.001: 5629 (99.9%)
. 0.001 to 0.1: 6 (0.1%)
. >= 0.1: 0 (0%)
. Median |diff|: 1e-04 | Max: 0.0057
. Agreement 'ALM_WT' vs 'ALM_WT_source' (n = 5636):
. < 0.001: 5635 (100%)
. 0.001 to 0.1: 1 (0%)
. >= 0.1: 0 (0%)
. Median |diff|: 0 | Max: 0.0023

-> 63 targets ..... [14.9s, 2+, 15-]

.. Split ALM by DXA method ..
-> 63 targets ..... [14.9s, 2+, 15-]

-> 63 targets ..... [14.9s, 2+, 15-]
. Method-specific ALM columns created:
. Hologic and Lunar splits for ALM, ALM_HT2, ALM_BMI, ALM_WT

-> 63 targets ..... [14.9s, 2+, 15-]

```

```

.. Derive Age ..
-> 63 targets ..... [14.9s, 2+, 15-]

-> 63 targets ..... [14.9s, 2+, 15-]
. Age derived from datbirth (Baseline, filled forward/back) and exam_date_iso.
. Total rows: 6114 | derived: 6114 | missing: 0
. Age range (non-missing): 50.2 - 91.5 years
. Mean ± SD: 68.6 ± 7.9 years
. Agreement with Age_source (rows with both non-missing, n = 6114):
. Difference < 0.1 yr (effectively identical): 6066 (99.2%)
. Difference 0.1-1 yr (minor rounding/timing): 47 (0.8%)
. Difference >= 1 yr (meaningful mismatch) : 1 (0%)
. Median absolute difference: 0.028 years
. Max absolute difference : 4.328 years
[11/20] deriving ...

-> 63 targets ..... [14.9s, 2+, 15-]

.. Derive BMI ..
-> 63 targets ..... [14.9s, 2+, 15-]

-> 63 targets ..... [14.9s, 2+, 15-]
. BMI derived from Weight and Height.
. Total rows: 6114 | derived: 6114 | missing: 0
. BMI range (non-missing): 15.1 - 45.5 kg/m²
. Mean ± SD: 25.9 ± 4.7 kg/m²
. Agreement with BMI_source (rows with both non-missing, n = 6112):
. Difference < 0.01 (effectively identical): 6099 (99.8%)
. Difference 0.01-0.5 (minor rounding): 13 (0.2%)
. Difference >= 0.5 (meaningful mismatch): 0 (0%)
. Median absolute difference: 0.002 kg/m²
. Max absolute difference : 0.17 kg/m²

-> 63 targets ..... [14.9s, 2+, 15-]

.. Deriving BMI category ..
-> 63 targets ..... [14.9s, 2+, 15-]

-> 63 targets ..... [14.9s, 2+, 15-]
. derive_bmi_category: BMI categories derived.
. Processed 6114 rows (0 missing BMI).
  Prevalence of Obesity: 18.5%
  Prevalence of Overweight + Obesity: 53.4%

-> 63 targets ..... [14.9s, 2+, 15-]

.. Derive ALM Indices ..
-> 63 targets ..... [14.9s, 2+, 15-]

-> 63 targets ..... [14.9s, 2+, 15-]

```

```

.. Raw indices (native DXA platform)
-> 63 targets ..... [14.9s, 2+, 15-]
. 'ALM': 6114 derived, 0 missing
. Range 9061.706 - 27682.402 |
  Mean  $\pm$  SD: 17015.358  $\pm$  2578.944
. 'ALM_HT2': 6114 derived, 0 missing
. Range 4.1 - 10.943 |
  Mean  $\pm$  SD: 6.577  $\pm$  0.857
. 'ALM_BMI': 6114 derived, 0 missing
. Range 0.362 - 1.054 |
  Mean  $\pm$  SD: 0.667  $\pm$  0.101
. 'ALM_WT': 6114 derived, 0 missing
. Range 0.168 - 0.381 |
  Mean  $\pm$  SD: 0.257  $\pm$  0.028

-> 63 targets ..... [14.9s, 2+, 15-]

.. Harmonised indices (Lunar converted to Hologic scale)
-> 63 targets ..... [14.9s, 2+, 15-]
. 'ALM_harmonised': 6114 derived, 0 missing
. Range 9061.706 - 27682.402 |
  Mean  $\pm$  SD: 16517.558  $\pm$  2609.432
. 'ALM_HT2_harmonised': 6114 derived, 0 missing
. Range 3.811 - 10.943 |
  Mean  $\pm$  SD: 6.384  $\pm$  0.871
. 'ALM_BMI_harmonised': 6114 derived, 0 missing
. Range 0.354 - 1.054 |
  Mean  $\pm$  SD: 0.647  $\pm$  0.101
. 'ALM_WT_harmonised': 6114 derived, 0 missing
. Range 0.168 - 0.381 |
  Mean  $\pm$  SD: 0.249  $\pm$  0.028
. Sanity check passed: ALM_harmonised == ALM for all 2824 Hologic rows.
. Agreement 'ALM' vs 'ALM_source' (n = 5636):
. < 0.001: 5629 (99.9%)
. 0.001 to 0.1: 0 (0%)
. >= 0.1: 7 (0.1%)
. Median |diff|: 0 | Max: 341.2286
7 row(s) differ by >= 0.1 between derived and source.
. Agreement 'ALM_HT2' vs 'ALM_HT2_source' (n = 5636):
. < 0.001: 5629 (99.9%)
. 0.001 to 0.1: 6 (0.1%)
. >= 0.1: 1 (0%)
. Median |diff|: 0 | Max: 0.1376
1 row(s) differ by >= 0.1 between derived and source.
. Agreement 'ALM_BMI' vs 'ALM_BMI_source' (n = 5635):
. < 0.001: 5629 (99.9%)
. 0.001 to 0.1: 6 (0.1%)
. >= 0.1: 0 (0%)
. Median |diff|: 1e-04 | Max: 0.0151
. Agreement 'ALM_WT' vs 'ALM_WT_source' (n = 5636):
. < 0.001: 5635 (100%)
. 0.001 to 0.1: 1 (0%)
. >= 0.1: 0 (0%)
. Median |diff|: 0 | Max: 0.006

-> 63 targets ..... [14.9s, 2+, 15-]

```

```

.. Split ALM by DXA method ..
-> 63 targets ..... [14.9s, 2+, 15-]

-> 63 targets ..... [14.9s, 2+, 15-]
. Method-specific ALM columns created:
. Hologic and Lunar splits for ALM, ALM_HT2, ALM_BMI, ALM_WT

-> 63 targets ..... [14.9s, 2+, 15-]

.. Derive Age ..
-> 63 targets ..... [14.9s, 2+, 15-]

-> 63 targets ..... [14.9s, 2+, 15-]
. Age derived from datbirth (Baseline, filled forward/back) and exam_date_iso.
. Total rows: 6114 | derived: 6114 | missing: 0
. Age range (non-missing): 50.2 - 91.5 years
. Mean ± SD: 68.6 ± 7.9 years
. Agreement with Age_source (rows with both non-missing, n = 6114):
. Difference < 0.1 yr (effectively identical): 6066 (99.2%)
. Difference 0.1-1 yr (minor rounding/timing): 47 (0.8%)
. Difference >= 1 yr (meaningful mismatch) : 1 (0%)
. Median absolute difference: 0.028 years
. Max absolute difference : 4.328 years
[12/20] deriving ...

-> 63 targets ..... [14.9s, 2+, 15-]

.. Derive BMI ..
-> 63 targets ..... [14.9s, 2+, 15-]

-> 63 targets ..... [14.9s, 2+, 15-]
. BMI derived from Weight and Height.
. Total rows: 6114 | derived: 6114 | missing: 0
. BMI range (non-missing): 15.1 - 45.5 kg/m²
. Mean ± SD: 25.9 ± 4.7 kg/m²
. Agreement with BMI_source (rows with both non-missing, n = 6112):
. Difference < 0.01 (effectively identical): 6099 (99.8%)
. Difference 0.01-0.5 (minor rounding): 13 (0.2%)
. Difference >= 0.5 (meaningful mismatch): 0 (0%)
. Median absolute difference: 0.002 kg/m²
. Max absolute difference : 0.17 kg/m²

-> 63 targets ..... [14.9s, 2+, 15-]

.. Deriving BMI category ..
-> 63 targets ..... [14.9s, 2+, 15-]

-> 63 targets ..... [14.9s, 2+, 15-]
. derive_bmi_category: BMI categories derived.
. Processed 6114 rows (0 missing BMI).

```

Prevalence of Obesity: 18.5%
Prevalence of Overweight + Obesity: 53.4%

```
-> 63 targets ..... [14.9s, 2+, 15-]

.. Derive ALM Indices ..
-> 63 targets ..... [14.9s, 2+, 15-]

-> 63 targets ..... [14.9s, 2+, 15-]

.. Raw indices (native DXA platform)
-> 63 targets ..... [14.9s, 2+, 15-]
. 'ALM': 6114 derived, 0 missing
. Range 9061.706 - 27682.402 |
  Mean ± SD: 17013.015 ± 2586.552
. 'ALM_HT2': 6114 derived, 0 missing
. Range 4.1 - 10.943 |
  Mean ± SD: 6.576 ± 0.858
. 'ALM_BMI': 6114 derived, 0 missing
. Range 0.379 - 1.054 |
  Mean ± SD: 0.667 ± 0.101
. 'ALM_WT': 6114 derived, 0 missing
. Range 0.175 - 0.381 |
  Mean ± SD: 0.257 ± 0.028

-> 63 targets ..... [14.9s, 2+, 15-]

.. Harmonised indices (Lunar converted to Hologic scale)
-> 63 targets ..... [14.9s, 2+, 15-]
. 'ALM_harmonised': 6114 derived, 0 missing
. Range 9061.706 - 27682.402 |
  Mean ± SD: 16515.184 ± 2616.196
. 'ALM_HT2_harmonised': 6114 derived, 0 missing
. Range 3.811 - 10.943 |
  Mean ± SD: 6.382 ± 0.872
. 'ALM_BMI_harmonised': 6114 derived, 0 missing
. Range 0.354 - 1.054 |
  Mean ± SD: 0.647 ± 0.101
. 'ALM_WT_harmonised': 6114 derived, 0 missing
. Range 0.168 - 0.381 |
  Mean ± SD: 0.249 ± 0.028
. Sanity check passed: ALM_harmonised == ALM for all 2824 Hologic rows.
. Agreement 'ALM' vs 'ALM_source' (n = 5636):
. < 0.001: 5629 (99.9%)
. 0.001 to 0.1: 0 (0%)
. >= 0.1: 7 (0.1%)
. Median |diff|: 0 | Max: 16.7426
7 row(s) differ by >= 0.1 between derived and source.
. Agreement 'ALM_HT2' vs 'ALM_HT2_source' (n = 5636):
. < 0.001: 5629 (99.9%)
. 0.001 to 0.1: 7 (0.1%)
. >= 0.1: 0 (0%)
. Median |diff|: 0 | Max: 0.0067
. Agreement 'ALM_BMI' vs 'ALM_BMI_source' (n = 5635):
. < 0.001: 5630 (99.9%)
```

```

. 0.001 to 0.1: 5 (0.1%)
. >= 0.1: 0 (0%)
. Median |diff|: 1e-04 | Max: 0.0052
. Agreement 'ALM_WT' vs 'ALM_WT_source' (n = 5636):
. < 0.001: 5636 (100%)
. 0.001 to 0.1: 0 (0%)
. >= 0.1: 0 (0%)
. Median |diff|: 0 | Max: 3e-04

-> 63 targets ..... [14.9s, 2+, 15-]

.. Split ALM by DXA method ..
-> 63 targets ..... [14.9s, 2+, 15-]

-> 63 targets ..... [14.9s, 2+, 15-]
. Method-specific ALM columns created:
. Hologic and Lunar splits for ALM, ALM_HT2, ALM_BMI, ALM_WT

-> 63 targets ..... [14.9s, 2+, 15-]

.. Derive Age ..
-> 63 targets ..... [14.9s, 2+, 15-]

-> 63 targets ..... [14.9s, 2+, 15-]
. Age derived from datbirth (Baseline, filled forward/back) and exam_date_iso.
. Total rows: 6114 | derived: 6114 | missing: 0
. Age range (non-missing): 50.2 - 91.5 years
. Mean ± SD: 68.6 ± 7.9 years
. Agreement with Age_source (rows with both non-missing, n = 6114):
. Difference < 0.1 yr (effectively identical): 6066 (99.2%)
. Difference 0.1-1 yr (minor rounding/timing): 47 (0.8%)
. Difference >= 1 yr (meaningful mismatch) : 1 (0%)
. Median absolute difference: 0.028 years
. Max absolute difference : 4.328 years
[13/20] deriving ...

-> 63 targets ..... [14.9s, 2+, 15-]

.. Derive BMI ..
-> 63 targets ..... [14.9s, 2+, 15-]

-> 63 targets ..... [14.9s, 2+, 15-]
. BMI derived from Weight and Height.
. Total rows: 6114 | derived: 6114 | missing: 0
. BMI range (non-missing): 15.1 - 45.5 kg/m²
. Mean ± SD: 25.9 ± 4.7 kg/m²
. Agreement with BMI_source (rows with both non-missing, n = 6112):
. Difference < 0.01 (effectively identical): 6099 (99.8%)
. Difference 0.01-0.5 (minor rounding): 13 (0.2%)
. Difference >= 0.5 (meaningful mismatch): 0 (0%)
. Median absolute difference: 0.002 kg/m²
. Max absolute difference : 0.17 kg/m²

```

```

-> 63 targets ..... [14.9s, 2+, 15-]

.. Deriving BMI category ..
-> 63 targets ..... [14.9s, 2+, 15-]

-> 63 targets ..... [14.9s, 2+, 15-]
. derive_bmi_category: BMI categories derived.
. Processed 6114 rows (0 missing BMI).
  Prevalence of Obesity: 18.5%
  Prevalence of Overweight + Obesity: 53.4%

-> 63 targets ..... [14.9s, 2+, 15-]

.. Derive ALM Indices ..
-> 63 targets ..... [14.9s, 2+, 15-]

-> 63 targets ..... [14.9s, 2+, 15-]

.. Raw indices (native DXA platform)
-> 63 targets ..... [14.9s, 2+, 15-]
. 'ALM': 6114 derived, 0 missing
. Range 9061.706 - 27823.745 |
  Mean ± SD: 17007.917 ± 2587.305
. 'ALM_HT2': 6114 derived, 0 missing
. Range 4.1 - 10.943 |
  Mean ± SD: 6.574 ± 0.859
. 'ALM_BMI': 6114 derived, 0 missing
. Range 0.379 - 1.054 |
  Mean ± SD: 0.666 ± 0.101
. 'ALM_WT': 6114 derived, 0 missing
. Range 0.178 - 0.381 |
  Mean ± SD: 0.257 ± 0.028

-> 63 targets ..... [14.9s, 2+, 15-]

.. Harmonised indices (Lunar converted to Hologic scale)
-> 63 targets ..... [14.9s, 2+, 15-]
. 'ALM_harmonised': 6114 derived, 0 missing
. Range 9061.706 - 27823.745 |
  Mean ± SD: 16510.068 ± 2615.76
. 'ALM_HT2_harmonised': 6114 derived, 0 missing
. Range 3.811 - 10.943 |
  Mean ± SD: 6.381 ± 0.872
. 'ALM_BMI_harmonised': 6114 derived, 0 missing
. Range 0.354 - 1.054 |
  Mean ± SD: 0.647 ± 0.101
. 'ALM_WT_harmonised': 6114 derived, 0 missing
. Range 0.168 - 0.381 |
  Mean ± SD: 0.249 ± 0.028
. Sanity check passed: ALM_harmonised == ALM for all 2824 Hologic rows.
. Agreement 'ALM' vs 'ALM_source' (n = 5636):
. < 0.001: 5629 (99.9%)

```

```

. 0.001 to 0.1: 0 (0%)
. >= 0.1: 7 (0.1%)
. Median |diff|: 0 | Max: 286.306
7 row(s) differ by >= 0.1 between derived and source.
. Agreement 'ALM_HT2' vs 'ALM_HT2_source' (n = 5636):
. < 0.001: 5629 (99.9%)
. 0.001 to 0.1: 6 (0.1%)
. >= 0.1: 1 (0%)
. Median |diff|: 0 | Max: 0.1154
1 row(s) differ by >= 0.1 between derived and source.
. Agreement 'ALM_BMI' vs 'ALM_BMI_source' (n = 5635):
. < 0.001: 5629 (99.9%)
. 0.001 to 0.1: 6 (0.1%)
. >= 0.1: 0 (0%)
. Median |diff|: 1e-04 | Max: 0.0124
. Agreement 'ALM_WT' vs 'ALM_WT_source' (n = 5636):
. < 0.001: 5635 (100%)
. 0.001 to 0.1: 1 (0%)
. >= 0.1: 0 (0%)
. Median |diff|: 0 | Max: 0.005

-> 63 targets ..... [14.9s, 2+, 15-]

.. Split ALM by DXA method ..
-> 63 targets ..... [14.9s, 2+, 15-]

-> 63 targets ..... [14.9s, 2+, 15-]
. Method-specific ALM columns created:
. Hologic and Lunar splits for ALM, ALM_HT2, ALM_BMI, ALM_WT

-> 63 targets ..... [14.9s, 2+, 15-]

.. Derive Age ..
-> 63 targets ..... [14.9s, 2+, 15-]

-> 63 targets ..... [14.9s, 2+, 15-]
. Age derived from datbirth (Baseline, filled forward/back) and exam_date_iso.
. Total rows: 6114 | derived: 6114 | missing: 0
. Age range (non-missing): 50.2 - 91.5 years
. Mean ± SD: 68.6 ± 7.9 years
. Agreement with Age_source (rows with both non-missing, n = 6114):
. Difference < 0.1 yr (effectively identical): 6066 (99.2%)
. Difference 0.1-1 yr (minor rounding/timing): 47 (0.8%)
. Difference >= 1 yr (meaningful mismatch) : 1 (0%)
. Median absolute difference: 0.028 years
. Max absolute difference : 4.328 years
[14/20] deriving ...

-> 63 targets ..... [14.9s, 2+, 15-]

.. Derive BMI ..
-> 63 targets ..... [14.9s, 2+, 15-]

```

```
-> 63 targets ..... [14.9s, 2+, 15-]
. BMI derived from Weight and Height.
. Total rows: 6114 | derived: 6114 | missing: 0
. BMI range (non-missing): 15.1 - 45.5 kg/m2
. Mean ± SD: 25.9 ± 4.7 kg/m2
. Agreement with BMI_source (rows with both non-missing, n = 6112):
. Difference < 0.01 (effectively identical): 6099 (99.8%)
. Difference 0.01-0.5 (minor rounding): 13 (0.2%)
. Difference >= 0.5 (meaningful mismatch): 0 (0%)
. Median absolute difference: 0.002 kg/m2
. Max absolute difference : 0.17 kg/m2
```

```
-> 63 targets ..... [14.9s, 2+, 15-]
```

```
.. Deriving BMI category ..
```

```
-> 63 targets ..... [14.9s, 2+, 15-]
```

```
-> 63 targets ..... [14.9s, 2+, 15-]
. derive_bmi_category: BMI categories derived.
. Processed 6114 rows (0 missing BMI).
  Prevalence of Obesity: 18.5%
  Prevalence of Overweight + Obesity: 53.4%
```

```
-> 63 targets ..... [14.9s, 2+, 15-]
```

```
.. Derive ALM Indices ..
```

```
-> 63 targets ..... [14.9s, 2+, 15-]
```

```
-> 63 targets ..... [14.9s, 2+, 15-]
```

```
.. Raw indices (native DXA platform)
```

```
-> 63 targets ..... [14.9s, 2+, 15-]
```

```
. 'ALM': 6114 derived, 0 missing
. Range 9061.706 - 27682.402 |
  Mean ± SD: 17008.244 ± 2585.945
. 'ALM_HT2': 6114 derived, 0 missing
. Range 4.1 - 10.943 |
  Mean ± SD: 6.574 ± 0.858
. 'ALM_BMI': 6114 derived, 0 missing
. Range 0.379 - 1.054 |
  Mean ± SD: 0.666 ± 0.101
. 'ALM_WT': 6114 derived, 0 missing
. Range 0.178 - 0.381 |
  Mean ± SD: 0.257 ± 0.028
```

```
-> 63 targets ..... [14.9s, 2+, 15-]
```

```
.. Harmonised indices (Lunar converted to Hologic scale)
```

```
-> 63 targets ..... [14.9s, 2+, 15-]
```

```
. 'ALM_harmonised': 6114 derived, 0 missing
. Range 9061.706 - 27682.402 |
  Mean ± SD: 16510.46 ± 2615.165
```

```

. 'ALM_HT2_harmonised': 6114 derived, 0 missing
. Range 3.811 - 10.943 |
  Mean ± SD: 6.381 ± 0.872
. 'ALM_BMI_harmonised': 6114 derived, 0 missing
. Range 0.354 - 1.054 |
  Mean ± SD: 0.647 ± 0.101
. 'ALM_WT_harmonised': 6114 derived, 0 missing
. Range 0.168 - 0.381 |
  Mean ± SD: 0.249 ± 0.028
. Sanity check passed: ALM_harmonised == ALM for all 2824 Hologic rows.
. Agreement 'ALM' vs 'ALM_source' (n = 5636):
. < 0.001: 5629 (99.9%)
. 0.001 to 0.1: 0 (0%)
. >= 0.1: 7 (0.1%)
. Median |diff|: 0 | Max: 84.0683
7 row(s) differ by >= 0.1 between derived and source.
. Agreement 'ALM_HT2' vs 'ALM_HT2_source' (n = 5636):
. < 0.001: 5629 (99.9%)
. 0.001 to 0.1: 7 (0.1%)
. >= 0.1: 0 (0%)
. Median |diff|: 0 | Max: 0.0339
. Agreement 'ALM_BMI' vs 'ALM_BMI_source' (n = 5635):
. < 0.001: 5629 (99.9%)
. 0.001 to 0.1: 6 (0.1%)
. >= 0.1: 0 (0%)
. Median |diff|: 1e-04 | Max: 0.0052
. Agreement 'ALM_WT' vs 'ALM_WT_source' (n = 5636):
. < 0.001: 5635 (100%)
. 0.001 to 0.1: 1 (0%)
. >= 0.1: 0 (0%)
. Median |diff|: 0 | Max: 0.0015

-> 63 targets ..... [14.9s, 2+, 15-]

.. Split ALM by DXA method ..
-> 63 targets ..... [14.9s, 2+, 15-]

-> 63 targets ..... [14.9s, 2+, 15-]
. Method-specific ALM columns created:
. Hologic and Lunar splits for ALM, ALM_HT2, ALM_BMI, ALM_WT

-> 63 targets ..... [14.9s, 2+, 15-]

.. Derive Age ..
-> 63 targets ..... [14.9s, 2+, 15-]

-> 63 targets ..... [14.9s, 2+, 15-]
. Age derived from datbirth (Baseline, filled forward/back) and exam_date_iso.
. Total rows: 6114 | derived: 6114 | missing: 0
. Age range (non-missing): 50.2 - 91.5 years
. Mean ± SD: 68.6 ± 7.9 years
. Agreement with Age_source (rows with both non-missing, n = 6114):
. Difference < 0.1 yr (effectively identical): 6066 (99.2%)
. Difference 0.1-1 yr (minor rounding/timing): 47 (0.8%)

```

```

. Difference >= 1 yr (meaningful mismatch) : 1 (0%)
. Median absolute difference: 0.028 years
. Max absolute difference : 4.328 years
[15/20] deriving ...

-> 63 targets ..... [14.9s, 2+, 15-]

.. Derive BMI ..
-> 63 targets ..... [14.9s, 2+, 15-]

-> 63 targets ..... [14.9s, 2+, 15-]
. BMI derived from Weight and Height.
. Total rows: 6114 | derived: 6114 | missing: 0
. BMI range (non-missing): 15.1 - 45.5 kg/m2
. Mean ± SD: 25.9 ± 4.7 kg/m2
. Agreement with BMI_source (rows with both non-missing, n = 6112):
. Difference < 0.01 (effectively identical): 6099 (99.8%)
. Difference 0.01-0.5 (minor rounding): 13 (0.2%)
. Difference >= 0.5 (meaningful mismatch): 0 (0%)
. Median absolute difference: 0.002 kg/m2
. Max absolute difference : 0.17 kg/m2

-> 63 targets ..... [14.9s, 2+, 15-]

.. Deriving BMI category ..
-> 63 targets ..... [14.9s, 2+, 15-]

-> 63 targets ..... [14.9s, 2+, 15-]
. derive_bmi_category: BMI categories derived.
. Processed 6114 rows (0 missing BMI).
  Prevalence of Obesity: 18.5%
  Prevalence of Overweight + Obesity: 53.4%

-> 63 targets ..... [14.9s, 2+, 15-]

.. Derive ALM Indices ..
-> 63 targets ..... [14.9s, 2+, 15-]

-> 63 targets ..... [14.9s, 2+, 15-]

.. Raw indices (native DXA platform)
-> 63 targets ..... [14.9s, 2+, 15-]
. 'ALM': 6114 derived, 0 missing
. Range 9061.706 - 27682.402 |
  Mean ± SD: 17006.247 ± 2587.621
. 'ALM_HT2': 6114 derived, 0 missing
. Range 4.1 - 10.943 |
  Mean ± SD: 6.573 ± 0.86
. 'ALM_BMI': 6114 derived, 0 missing
. Range 0.379 - 1.054 |
  Mean ± SD: 0.666 ± 0.101
. 'ALM_WT': 6114 derived, 0 missing

```

```
. Range 0.178 - 0.381 |
Mean ± SD: 0.257 ± 0.028
```

```
-> 63 targets ..... [14.9s, 2+, 15-]
```

```
.. Harmonised indices (Lunar converted to Hologic scale)
```

```
-> 63 targets ..... [14.9s, 2+, 15-]
```

```
. 'ALM_harmonised': 6114 derived, 0 missing
. Range 9061.706 - 27682.402 |
Mean ± SD: 16508.451 ± 2616.329
. 'ALM_HT2_harmonised': 6114 derived, 0 missing
. Range 3.811 - 10.943 |
Mean ± SD: 6.38 ± 0.873
. 'ALM_BMI_harmonised': 6114 derived, 0 missing
. Range 0.354 - 1.054 |
Mean ± SD: 0.647 ± 0.101
. 'ALM_WT_harmonised': 6114 derived, 0 missing
. Range 0.168 - 0.381 |
Mean ± SD: 0.249 ± 0.028
. Sanity check passed: ALM_harmonised == ALM for all 2824 Hologic rows.
. Agreement 'ALM' vs 'ALM_source' (n = 5636):
. < 0.001: 5629 (99.9%)
. 0.001 to 0.1: 0 (0%)
. >= 0.1: 7 (0.1%)
. Median |diff|: 0 | Max: 267.3757
7 row(s) differ by >= 0.1 between derived and source.
. Agreement 'ALM_HT2' vs 'ALM_HT2_source' (n = 5636):
. < 0.001: 5629 (99.9%)
. 0.001 to 0.1: 6 (0.1%)
. >= 0.1: 1 (0%)
. Median |diff|: 0 | Max: 0.1078
1 row(s) differ by >= 0.1 between derived and source.
. Agreement 'ALM_BMI' vs 'ALM_BMI_source' (n = 5635):
. < 0.001: 5629 (99.9%)
. 0.001 to 0.1: 6 (0.1%)
. >= 0.1: 0 (0%)
. Median |diff|: 1e-04 | Max: 0.0118
. Agreement 'ALM_WT' vs 'ALM_WT_source' (n = 5636):
. < 0.001: 5635 (100%)
. 0.001 to 0.1: 1 (0%)
. >= 0.1: 0 (0%)
. Median |diff|: 0 | Max: 0.0047
```

```
-> 63 targets ..... [14.9s, 2+, 15-]
```

```
.. Split ALM by DXA method ..
```

```
-> 63 targets ..... [14.9s, 2+, 15-]
```

```
-> 63 targets ..... [14.9s, 2+, 15-]
```

```
. Method-specific ALM columns created:
. Hologic and Lunar splits for ALM, ALM_HT2, ALM_BMI, ALM_WT
```

```
-> 63 targets ..... [14.9s, 2+, 15-]
```

```

.. Derive Age ..
-> 63 targets ..... [14.9s, 2+, 15-]

-> 63 targets ..... [14.9s, 2+, 15-]
. Age derived from datbirth (Baseline, filled forward/back) and exam_date_iso.
. Total rows: 6114 | derived: 6114 | missing: 0
. Age range (non-missing): 50.2 - 91.5 years
. Mean ± SD: 68.6 ± 7.9 years
. Agreement with Age_source (rows with both non-missing, n = 6114):
. Difference < 0.1 yr (effectively identical): 6066 (99.2%)
. Difference 0.1-1 yr (minor rounding/timing): 47 (0.8%)
. Difference >= 1 yr (meaningful mismatch) : 1 (0%)
. Median absolute difference: 0.028 years
. Max absolute difference : 4.328 years
[16/20] deriving ...

-> 63 targets ..... [14.9s, 2+, 15-]

.. Derive BMI ..
-> 63 targets ..... [14.9s, 2+, 15-]

-> 63 targets ..... [14.9s, 2+, 15-]
. BMI derived from Weight and Height.
. Total rows: 6114 | derived: 6114 | missing: 0
. BMI range (non-missing): 15.1 - 45.5 kg/m²
. Mean ± SD: 25.9 ± 4.7 kg/m²
. Agreement with BMI_source (rows with both non-missing, n = 6112):
. Difference < 0.01 (effectively identical): 6099 (99.8%)
. Difference 0.01-0.5 (minor rounding): 13 (0.2%)
. Difference >= 0.5 (meaningful mismatch): 0 (0%)
. Median absolute difference: 0.002 kg/m²
. Max absolute difference : 0.17 kg/m²

-> 63 targets ..... [14.9s, 2+, 15-]

.. Deriving BMI category ..
-> 63 targets ..... [14.9s, 2+, 15-]

-> 63 targets ..... [14.9s, 2+, 15-]
. derive_bmi_category: BMI categories derived.
. Processed 6114 rows (0 missing BMI).
  Prevalence of Obesity: 18.5%
  Prevalence of Overweight + Obesity: 53.4%

-> 63 targets ..... [14.9s, 2+, 15-]

.. Derive ALM Indices ..
-> 63 targets ..... [14.9s, 2+, 15-]

-> 63 targets ..... [14.9s, 2+, 15-]

```

```

.. Raw indices (native DXA platform)
-> 63 targets ..... [14.9s, 2+, 15-]
. 'ALM': 6114 derived, 0 missing
. Range 9061.706 - 27682.402 |
  Mean ± SD: 17005.906 ± 2591.493
. 'ALM_HT2': 6114 derived, 0 missing
. Range 4.1 - 10.943 |
  Mean ± SD: 6.573 ± 0.859
. 'ALM_BMI': 6114 derived, 0 missing
. Range 0.379 - 1.054 |
  Mean ± SD: 0.666 ± 0.101
. 'ALM_WT': 6114 derived, 0 missing
. Range 0.176 - 0.381 |
  Mean ± SD: 0.257 ± 0.028

-> 63 targets ..... [14.9s, 2+, 15-]

.. Harmonised indices (Lunar converted to Hologic scale)
-> 63 targets ..... [14.9s, 2+, 15-]
. 'ALM_harmonised': 6114 derived, 0 missing
. Range 9061.706 - 27682.402 |
  Mean ± SD: 16508.112 ± 2620.099
. 'ALM_HT2_harmonised': 6114 derived, 0 missing
. Range 3.811 - 10.943 |
  Mean ± SD: 6.379 ± 0.872
. 'ALM_BMI_harmonised': 6114 derived, 0 missing
. Range 0.354 - 1.054 |
  Mean ± SD: 0.647 ± 0.101
. 'ALM_WT_harmonised': 6114 derived, 0 missing
. Range 0.168 - 0.381 |
  Mean ± SD: 0.249 ± 0.028
. Sanity check passed: ALM_harmonised == ALM for all 2824 Hologic rows.
. Agreement 'ALM' vs 'ALM_source' (n = 5636):
. < 0.001: 5629 (99.9%)
. 0.001 to 0.1: 0 (0%)
. >= 0.1: 7 (0.1%)
. Median |diff|: 0 | Max: 194.921
7 row(s) differ by >= 0.1 between derived and source.
. Agreement 'ALM_HT2' vs 'ALM_HT2_source' (n = 5636):
. < 0.001: 5629 (99.9%)
. 0.001 to 0.1: 7 (0.1%)
. >= 0.1: 0 (0%)
. Median |diff|: 0 | Max: 0.0786
. Agreement 'ALM_BMI' vs 'ALM_BMI_source' (n = 5635):
. < 0.001: 5629 (99.9%)
. 0.001 to 0.1: 6 (0.1%)
. >= 0.1: 0 (0%)
. Median |diff|: 1e-04 | Max: 0.0087
. Agreement 'ALM_WT' vs 'ALM_WT_source' (n = 5636):
. < 0.001: 5635 (100%)
. 0.001 to 0.1: 1 (0%)
. >= 0.1: 0 (0%)
. Median |diff|: 0 | Max: 0.0035

-> 63 targets ..... [14.9s, 2+, 15-]

```

```

.. Split ALM by DXA method ..
-> 63 targets ..... [14.9s, 2+, 15-]

-> 63 targets ..... [14.9s, 2+, 15-]
. Method-specific ALM columns created:
. Hologic and Lunar splits for ALM, ALM_HT2, ALM_BMI, ALM_WT

-> 63 targets ..... [14.9s, 2+, 15-]

.. Derive Age ..
-> 63 targets ..... [14.9s, 2+, 15-]

-> 63 targets ..... [14.9s, 2+, 15-]
. Age derived from datbirth (Baseline, filled forward/back) and exam_date_iso.
. Total rows: 6114 | derived: 6114 | missing: 0
. Age range (non-missing): 50.2 - 91.5 years
. Mean ± SD: 68.6 ± 7.9 years
. Agreement with Age_source (rows with both non-missing, n = 6114):
. Difference < 0.1 yr (effectively identical): 6066 (99.2%)
. Difference 0.1-1 yr (minor rounding/timing): 47 (0.8%)
. Difference >= 1 yr (meaningful mismatch) : 1 (0%)
. Median absolute difference: 0.028 years
. Max absolute difference : 4.328 years
[17/20] deriving ...

-> 63 targets ..... [14.9s, 2+, 15-]

.. Derive BMI ..
-> 63 targets ..... [14.9s, 2+, 15-]

-> 63 targets ..... [14.9s, 2+, 15-]
. BMI derived from Weight and Height.
. Total rows: 6114 | derived: 6114 | missing: 0
. BMI range (non-missing): 15.1 - 45.5 kg/m²
. Mean ± SD: 25.9 ± 4.7 kg/m²
. Agreement with BMI_source (rows with both non-missing, n = 6112):
. Difference < 0.01 (effectively identical): 6099 (99.8%)
. Difference 0.01-0.5 (minor rounding): 13 (0.2%)
. Difference >= 0.5 (meaningful mismatch): 0 (0%)
. Median absolute difference: 0.002 kg/m²
. Max absolute difference : 0.17 kg/m²

-> 63 targets ..... [14.9s, 2+, 15-]

.. Deriving BMI category ..
-> 63 targets ..... [14.9s, 2+, 15-]

-> 63 targets ..... [14.9s, 2+, 15-]
. derive_bmi_category: BMI categories derived.
. Processed 6114 rows (0 missing BMI).
  Prevalence of Obesity: 18.5%

```

Prevalence of Overweight + Obesity: 53.4%

```
-> 63 targets ..... [14.9s, 2+, 15-]

.. Derive ALM Indices ..
-> 63 targets ..... [14.9s, 2+, 15-]

-> 63 targets ..... [14.9s, 2+, 15-]

.. Raw indices (native DXA platform)
-> 63 targets ..... [14.9s, 2+, 15-]
. 'ALM': 6114 derived, 0 missing
. Range 9061.706 - 27682.402 |
  Mean ± SD: 17004.137 ± 2583.394
. 'ALM_HT2': 6114 derived, 0 missing
. Range 4.1 - 10.943 |
  Mean ± SD: 6.573 ± 0.857
. 'ALM_BMI': 6114 derived, 0 missing
. Range 0.379 - 1.054 |
  Mean ± SD: 0.666 ± 0.101
. 'ALM_WT': 6114 derived, 0 missing
. Range 0.178 - 0.381 |
  Mean ± SD: 0.257 ± 0.028

-> 63 targets ..... [14.9s, 2+, 15-]

.. Harmonised indices (Lunar converted to Hologic scale)
-> 63 targets ..... [14.9s, 2+, 15-]
. 'ALM_harmonised': 6114 derived, 0 missing
. Range 9061.706 - 27682.402 |
  Mean ± SD: 16506.341 ± 2611.726
. 'ALM_HT2_harmonised': 6114 derived, 0 missing
. Range 3.811 - 10.943 |
  Mean ± SD: 6.379 ± 0.869
. 'ALM_BMI_harmonised': 6114 derived, 0 missing
. Range 0.354 - 1.054 |
  Mean ± SD: 0.647 ± 0.101
. 'ALM_WT_harmonised': 6114 derived, 0 missing
. Range 0.168 - 0.381 |
  Mean ± SD: 0.249 ± 0.029
. Sanity check passed: ALM_harmonised == ALM for all 2824 Hologic rows.
. Agreement 'ALM' vs 'ALM_source' (n = 5636):
. < 0.001: 5629 (99.9%)
. 0.001 to 0.1: 0 (0%)
. >= 0.1: 7 (0.1%)
. Median |diff|: 0 | Max: 72.5817
7 row(s) differ by >= 0.1 between derived and source.
. Agreement 'ALM_HT2' vs 'ALM_HT2_source' (n = 5636):
. < 0.001: 5629 (99.9%)
. 0.001 to 0.1: 7 (0.1%)
. >= 0.1: 0 (0%)
. Median |diff|: 0 | Max: 0.0293
. Agreement 'ALM_BMI' vs 'ALM_BMI_source' (n = 5635):
. < 0.001: 5629 (99.9%)
. 0.001 to 0.1: 6 (0.1%)
```

```

. >= 0.1: 0 (0%)
. Median |diff|: 1e-04 | Max: 0.0052
. Agreement 'ALM_WT' vs 'ALM_WT_source' (n = 5636):
. < 0.001: 5635 (100%)
. 0.001 to 0.1: 1 (0%)
. >= 0.1: 0 (0%)
. Median |diff|: 0 | Max: 0.0013

-> 63 targets ..... [14.9s, 2+, 15-]

.. Split ALM by DXA method ..
-> 63 targets ..... [14.9s, 2+, 15-]

-> 63 targets ..... [14.9s, 2+, 15-]
. Method-specific ALM columns created:
. Hologic and Lunar splits for ALM, ALM_HT2, ALM_BMI, ALM_WT

-> 63 targets ..... [14.9s, 2+, 15-]

.. Derive Age ..
-> 63 targets ..... [14.9s, 2+, 15-]

-> 63 targets ..... [14.9s, 2+, 15-]
. Age derived from datbirth (Baseline, filled forward/back) and exam_date_iso.
. Total rows: 6114 | derived: 6114 | missing: 0
. Age range (non-missing): 50.2 - 91.5 years
. Mean ± SD: 68.6 ± 7.9 years
. Agreement with Age_source (rows with both non-missing, n = 6114):
. Difference < 0.1 yr (effectively identical): 6066 (99.2%)
. Difference 0.1-1 yr (minor rounding/timing): 47 (0.8%)
. Difference >= 1 yr (meaningful mismatch) : 1 (0%)
. Median absolute difference: 0.028 years
. Max absolute difference : 4.328 years
[18/20] deriving ...

-> 63 targets ..... [14.9s, 2+, 15-]

.. Derive BMI ..
-> 63 targets ..... [14.9s, 2+, 15-]

-> 63 targets ..... [14.9s, 2+, 15-]
. BMI derived from Weight and Height.
. Total rows: 6114 | derived: 6114 | missing: 0
. BMI range (non-missing): 15.1 - 45.5 kg/m²
. Mean ± SD: 25.9 ± 4.7 kg/m²
. Agreement with BMI_source (rows with both non-missing, n = 6112):
. Difference < 0.01 (effectively identical): 6099 (99.8%)
. Difference 0.01-0.5 (minor rounding): 13 (0.2%)
. Difference >= 0.5 (meaningful mismatch): 0 (0%)
. Median absolute difference: 0.002 kg/m²
. Max absolute difference : 0.17 kg/m²

```

```

-> 63 targets ..... [14.9s, 2+, 15-]

.. Deriving BMI category ..
-> 63 targets ..... [14.9s, 2+, 15-]

-> 63 targets ..... [14.9s, 2+, 15-]
. derive_bmi_category: BMI categories derived.
. Processed 6114 rows (0 missing BMI).
  Prevalence of Obesity: 18.5%
  Prevalence of Overweight + Obesity: 53.4%

-> 63 targets ..... [14.9s, 2+, 15-]

.. Derive ALM Indices ..
-> 63 targets ..... [14.9s, 2+, 15-]

-> 63 targets ..... [14.9s, 2+, 15-]

.. Raw indices (native DXA platform)
-> 63 targets ..... [14.9s, 2+, 15-]
. 'ALM': 6114 derived, 0 missing
. Range 9061.706 - 27682.402 |
  Mean ± SD: 17002.561 ± 2592.547
. 'ALM_HT2': 6114 derived, 0 missing
. Range 4.071 - 10.943 |
  Mean ± SD: 6.572 ± 0.861
. 'ALM_BMI': 6114 derived, 0 missing
. Range 0.379 - 1.054 |
  Mean ± SD: 0.666 ± 0.101
. 'ALM_WT': 6114 derived, 0 missing
. Range 0.178 - 0.381 |
  Mean ± SD: 0.257 ± 0.028

-> 63 targets ..... [14.9s, 2+, 15-]

.. Harmonised indices (Lunar converted to Hologic scale)
-> 63 targets ..... [14.9s, 2+, 15-]
. 'ALM_harmonised': 6114 derived, 0 missing
. Range 9061.706 - 27682.402 |
  Mean ± SD: 16504.781 ± 2620.656
. 'ALM_HT2_harmonised': 6114 derived, 0 missing
. Range 3.811 - 10.943 |
  Mean ± SD: 6.378 ± 0.874
. 'ALM_BMI_harmonised': 6114 derived, 0 missing
. Range 0.354 - 1.054 |
  Mean ± SD: 0.646 ± 0.101
. 'ALM_WT_harmonised': 6114 derived, 0 missing
. Range 0.168 - 0.381 |
  Mean ± SD: 0.249 ± 0.028
. Sanity check passed: ALM_harmonised == ALM for all 2824 Hologic rows.
. Agreement 'ALM' vs 'ALM_source' (n = 5636):
. < 0.001: 5629 (99.9%)
. 0.001 to 0.1: 0 (0%)

```

```

. >= 0.1: 7 (0.1%)
. Median |diff|: 0 | Max: 239.5413
7 row(s) differ by >= 0.1 between derived and source.
. Agreement 'ALM_HT2' vs 'ALM_HT2_source' (n = 5636):
. < 0.001: 5629 (99.9%)
. 0.001 to 0.1: 7 (0.1%)
. >= 0.1: 0 (0%)
. Median |diff|: 0 | Max: 0.0966
. Agreement 'ALM_BMI' vs 'ALM_BMI_source' (n = 5635):
. < 0.001: 5629 (99.9%)
. 0.001 to 0.1: 6 (0.1%)
. >= 0.1: 0 (0%)
. Median |diff|: 1e-04 | Max: 0.0106
. Agreement 'ALM_WT' vs 'ALM_WT_source' (n = 5636):
. < 0.001: 5635 (100%)
. 0.001 to 0.1: 1 (0%)
. >= 0.1: 0 (0%)
. Median |diff|: 0 | Max: 0.0042

-> 63 targets ..... [14.9s, 2+, 15-]

.. Split ALM by DXA method ..
-> 63 targets ..... [14.9s, 2+, 15-]

-> 63 targets ..... [14.9s, 2+, 15-]
. Method-specific ALM columns created:
. Hologic and Lunar splits for ALM, ALM_HT2, ALM_BMI, ALM_WT

-> 63 targets ..... [14.9s, 2+, 15-]

.. Derive Age ..
-> 63 targets ..... [14.9s, 2+, 15-]

-> 63 targets ..... [14.9s, 2+, 15-]
. Age derived from datbirth (Baseline, filled forward/back) and exam_date_iso.
. Total rows: 6114 | derived: 6114 | missing: 0
. Age range (non-missing): 50.2 - 91.5 years
. Mean ± SD: 68.6 ± 7.9 years
. Agreement with Age_source (rows with both non-missing, n = 6114):
. Difference < 0.1 yr (effectively identical): 6066 (99.2%)
. Difference 0.1-1 yr (minor rounding/timing): 47 (0.8%)
. Difference >= 1 yr (meaningful mismatch) : 1 (0%)
. Median absolute difference: 0.028 years
. Max absolute difference : 4.328 years
[19/20] deriving ...

-> 63 targets ..... [14.9s, 2+, 15-]

.. Derive BMI ..
-> 63 targets ..... [14.9s, 2+, 15-]

-> 63 targets ..... [14.9s, 2+, 15-]

```

```

. BMI derived from Weight and Height.
. Total rows: 6114 | derived: 6114 | missing: 0
. BMI range (non-missing): 15.1 - 45.5 kg/m²
. Mean ± SD: 25.9 ± 4.7 kg/m²
. Agreement with BMI_source (rows with both non-missing, n = 6112):
. Difference < 0.01 (effectively identical): 6099 (99.8%)
. Difference 0.01-0.5 (minor rounding): 13 (0.2%)
. Difference >= 0.5 (meaningful mismatch): 0 (0%)
. Median absolute difference: 0.002 kg/m²
. Max absolute difference : 0.17 kg/m²

```

```

-> 63 targets ..... [14.9s, 2+, 15-]

```

```

.. Deriving BMI category ..

```

```

-> 63 targets ..... [14.9s, 2+, 15-]

```

```

-> 63 targets ..... [14.9s, 2+, 15-]

```

```

. derive_bmi_category: BMI categories derived.
. Processed 6114 rows (0 missing BMI).
  Prevalence of Obesity: 18.5%
  Prevalence of Overweight + Obesity: 53.4%

```

```

-> 63 targets ..... [14.9s, 2+, 15-]

```

```

.. Derive ALM Indices ..

```

```

-> 63 targets ..... [14.9s, 2+, 15-]

```

```

-> 63 targets ..... [14.9s, 2+, 15-]

```

```

.. Raw indices (native DXA platform)

```

```

-> 63 targets ..... [14.9s, 2+, 15-]

```

```

. 'ALM': 6114 derived, 0 missing
. Range 9061.706 - 27682.402 |
  Mean ± SD: 17002.721 ± 2593.035
. 'ALM_HT2': 6114 derived, 0 missing
. Range 4.1 - 10.943 |
  Mean ± SD: 6.572 ± 0.861
. 'ALM_BMI': 6114 derived, 0 missing
. Range 0.379 - 1.054 |
  Mean ± SD: 0.666 ± 0.101
. 'ALM_WT': 6114 derived, 0 missing
. Range 0.175 - 0.381 |
  Mean ± SD: 0.257 ± 0.028

```

```

-> 63 targets ..... [14.9s, 2+, 15-]

```

```

.. Harmonised indices (Lunar converted to Hologic scale)

```

```

-> 63 targets ..... [14.9s, 2+, 15-]

```

```

. 'ALM_harmonised': 6114 derived, 0 missing
. Range 9061.706 - 27682.402 |
  Mean ± SD: 16504.911 ± 2620.87
. 'ALM_HT2_harmonised': 6114 derived, 0 missing
. Range 3.811 - 10.943 |

```

```

Mean ± SD: 6.378 ± 0.873
. 'ALM_BMI_harmonised': 6114 derived, 0 missing
. Range 0.354 - 1.054 |
  Mean ± SD: 0.646 ± 0.101
. 'ALM_WT_harmonised': 6114 derived, 0 missing
. Range 0.168 - 0.381 |
  Mean ± SD: 0.249 ± 0.028
. Sanity check passed: ALM_harmonised == ALM for all 2824 Hologic rows.
. Agreement 'ALM' vs 'ALM_source' (n = 5636):
. < 0.001: 5629 (99.9%)
. 0.001 to 0.1: 0 (0%)
. >= 0.1: 7 (0.1%)
. Median |diff|: 0 | Max: 12.636
7 row(s) differ by >= 0.1 between derived and source.
. Agreement 'ALM_HT2' vs 'ALM_HT2_source' (n = 5636):
. < 0.001: 5629 (99.9%)
. 0.001 to 0.1: 7 (0.1%)
. >= 0.1: 0 (0%)
. Median |diff|: 0 | Max: 0.0051
. Agreement 'ALM_BMI' vs 'ALM_BMI_source' (n = 5635):
. < 0.001: 5630 (99.9%)
. 0.001 to 0.1: 5 (0.1%)
. >= 0.1: 0 (0%)
. Median |diff|: 1e-04 | Max: 0.0052
. Agreement 'ALM_WT' vs 'ALM_WT_source' (n = 5636):
. < 0.001: 5636 (100%)
. 0.001 to 0.1: 0 (0%)
. >= 0.1: 0 (0%)
. Median |diff|: 0 | Max: 2e-04

-> 63 targets ..... [14.9s, 2+, 15-]

.. Split ALM by DXA method ..
-> 63 targets ..... [14.9s, 2+, 15-]

-> 63 targets ..... [14.9s, 2+, 15-]
. Method-specific ALM columns created:
. Hologic and Lunar splits for ALM, ALM_HT2, ALM_BMI, ALM_WT

-> 63 targets ..... [14.9s, 2+, 15-]

.. Derive Age ..
-> 63 targets ..... [14.9s, 2+, 15-]

-> 63 targets ..... [14.9s, 2+, 15-]
. Age derived from datbirth (Baseline, filled forward/back) and exam_date_iso.
. Total rows: 6114 | derived: 6114 | missing: 0
. Age range (non-missing): 50.2 - 91.5 years
. Mean ± SD: 68.6 ± 7.9 years
. Agreement with Age_source (rows with both non-missing, n = 6114):
. Difference < 0.1 yr (effectively identical): 6066 (99.2%)
. Difference 0.1-1 yr (minor rounding/timing): 47 (0.8%)
. Difference >= 1 yr (meaningful mismatch) : 1 (0%)
. Median absolute difference: 0.028 years

```

. Max absolute difference : 4.328 years
[20/20] deriving ...

-> 63 targets [14.9s, 2+, 15-]

.. Derive BMI ..

-> 63 targets [14.9s, 2+, 15-]

-> 63 targets [14.9s, 2+, 15-]
. BMI derived from Weight and Height.
. Total rows: 6114 | derived: 6114 | missing: 0
. BMI range (non-missing): 15.1 - 45.5 kg/m²
. Mean ± SD: 25.9 ± 4.7 kg/m²
. Agreement with BMI_source (rows with both non-missing, n = 6112):
. Difference < 0.01 (effectively identical): 6099 (99.8%)
. Difference 0.01-0.5 (minor rounding): 13 (0.2%)
. Difference >= 0.5 (meaningful mismatch): 0 (0%)
. Median absolute difference: 0.002 kg/m²
. Max absolute difference : 0.17 kg/m²

-> 63 targets [14.9s, 2+, 15-]

.. Deriving BMI category ..

-> 63 targets [14.9s, 2+, 15-]

-> 63 targets [14.9s, 2+, 15-]
. derive_bmi_category: BMI categories derived.
. Processed 6114 rows (0 missing BMI).
Prevalence of Obesity: 18.5%
Prevalence of Overweight + Obesity: 53.4%

-> 63 targets [14.9s, 2+, 15-]

.. Derive ALM Indices ..

-> 63 targets [14.9s, 2+, 15-]

-> 63 targets [14.9s, 2+, 15-]

.. Raw indices (native DXA platform)

-> 63 targets [14.9s, 2+, 15-]
. 'ALM': 6114 derived, 0 missing
. Range 9061.706 - 27682.402 |
Mean ± SD: 17008.298 ± 2585.76
. 'ALM_HT2': 6114 derived, 0 missing
. Range 4.1 - 10.943 |
Mean ± SD: 6.574 ± 0.858
. 'ALM_BMI': 6114 derived, 0 missing
. Range 0.379 - 1.054 |
Mean ± SD: 0.666 ± 0.101
. 'ALM_WT': 6114 derived, 0 missing
. Range 0.178 - 0.381 |
Mean ± SD: 0.257 ± 0.028

```

-> 63 targets ..... [14.9s, 2+, 15-]

.. Harmonised indices (Lunar converted to Hologic scale)
-> 63 targets ..... [14.9s, 2+, 15-]
. 'ALM_harmonised': 6114 derived, 0 missing
. Range 9061.706 - 27682.402 |
  Mean ± SD: 16510.489 ± 2614.758
. 'ALM_HT2_harmonised': 6114 derived, 0 missing
. Range 3.811 - 10.943 |
  Mean ± SD: 6.381 ± 0.871
. 'ALM_BMI_harmonised': 6114 derived, 0 missing
. Range 0.354 - 1.054 |
  Mean ± SD: 0.647 ± 0.101
. 'ALM_WT_harmonised': 6114 derived, 0 missing
. Range 0.168 - 0.381 |
  Mean ± SD: 0.249 ± 0.028
. Sanity check passed: ALM_harmonised == ALM for all 2824 Hologic rows.
. Agreement 'ALM' vs 'ALM_source' (n = 5636):
. < 0.001: 5629 (99.9%)
. 0.001 to 0.1: 0 (0%)
. >= 0.1: 7 (0.1%)
. Median |diff|: 0 | Max: 83.7156
7 row(s) differ by >= 0.1 between derived and source.
. Agreement 'ALM_HT2' vs 'ALM_HT2_source' (n = 5636):
. < 0.001: 5629 (99.9%)
. 0.001 to 0.1: 7 (0.1%)
. >= 0.1: 0 (0%)
. Median |diff|: 0 | Max: 0.0338
. Agreement 'ALM_BMI' vs 'ALM_BMI_source' (n = 5635):
. < 0.001: 5629 (99.9%)
. 0.001 to 0.1: 6 (0.1%)
. >= 0.1: 0 (0%)
. Median |diff|: 1e-04 | Max: 0.0052
. Agreement 'ALM_WT' vs 'ALM_WT_source' (n = 5636):
. < 0.001: 5635 (100%)
. 0.001 to 0.1: 1 (0%)
. >= 0.1: 0 (0%)
. Median |diff|: 0 | Max: 0.0015

-> 63 targets ..... [14.9s, 2+, 15-]

.. Split ALM by DXA method ..
-> 63 targets ..... [14.9s, 2+, 15-]

-> 63 targets ..... [14.9s, 2+, 15-]
. Method-specific ALM columns created:
. Hologic and Lunar splits for ALM, ALM_HT2, ALM_BMI, ALM_WT

-> 63 targets ..... [14.9s, 2+, 15-]

.. Derive Age ..
-> 63 targets ..... [14.9s, 2+, 15-]

```

```
-> 63 targets ..... [14.9s, 2+, 15-]
. Age derived from datbirth (Baseline, filled forward/back) and exam_date_iso.
. Total rows: 6114 | derived: 6114 | missing: 0
. Age range (non-missing): 50.2 - 91.5 years
. Mean  $\pm$  SD: 68.6  $\pm$  7.9 years
. Agreement with Age_source (rows with both non-missing, n = 6114):
. Difference < 0.1 yr (effectively identical): 6066 (99.2%)
. Difference 0.1-1 yr (minor rounding/timing): 47 (0.8%)
. Difference >= 1 yr (meaningful mismatch) : 1 (0%)
. Median absolute difference: 0.028 years
. Max absolute difference : 4.328 years
. derive() complete.
. 122280 rows across 20 imputed datasets.
```